# Urban Economic Structure and Disaster Resilience
## The Role of Labor Markets, Road Networks, and Trade Linkages

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2026-09-08

**Abstract**

Agglomeration generates economic benefits by concentrating economic activity, but whether these same urban structures amplify or mitigate vulnerability to natural disasters remains unclear. I examine how industrial composition, road networks, and trade linkages shape disaster resilience across U.S. metropolitan areas. Natural disasters reduce local economic activity by approximately 3 percent on average. Valuing the associated loss in electricity service, the median estimated economic loss is equivalent to approximately 14 percent of monthly MSA GDP. The effect varies substantially with urban structure. Greater industrial concentration more than doubles the decline, whereas more centralized road networks and stronger intra-city goods sourcing reduce it by around half. The effects of industrial concentration also depend on where that concentration occurs, with concentration in Manufacturing mitigating disaster impacts and concentration in Accommodation, Wholesale Trade, and Transportation and in Health and Education amplifying them. These findings show that urban structures associated with agglomeration can either amplify or mitigate disaster impacts, making resilience an important dimension of how the benefits and costs of agglomeration are evaluated.

# I Introduction

Cities concentrate economic activity within a confined space. In the United States, metropolitan statistical areas (MSAs) account for the vast majority of employer establishments and population[1],

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[1] In 2023, MSAs contained approximately 7.9 million of the nation's 8.4 million employer establishments, accounting for nearly 95 percent of all employer establishments in the United States. Likewise, approximately 86 percent of the U.S. population, 294 million people, resided in MSAs as of July 1, 2024.

making cities the primary locus of production, employment, and innovation. This geographic concentration gives rise to agglomeration economies and the productivity gains they generate. Yet while these benefits are well established, much less is known about how the economic structures underlying them shape cities' resilience when disruptions occur. This question is particularly important in the context of natural disasters, as greater geographic concentration of economic activity creates the potential for greater economic losses when disruptions occur. Its importance is growing as natural disasters become more frequent and severe[2]. Understanding urban agglomeration therefore requires considering not only the productivity benefits it generates under normal conditions, but also how the underlying organization of economic activity shapes resilience to disruptive shocks.

When considering agglomeration in the context of natural disasters, an important insight from the literature is that agglomeration gains are multidimensional. They arise through multiple dimensions of urban economic structure, including employment composition, road networks, and trade linkages[3]. How these dimensions are configured shapes the agglomeration benefits a city realizes in normal times. Yet much less is known about whether the same multidimensional structures systematically shape economic resilience when natural disasters occur. For instance, the composition of employment determines which industries are most exposed when a disaster occurs, so greater concentration in particularly sensitive industries may generate larger losses. The organization of road networks may similarly matter for recovery, as a more concentrated network could facilitate targeted restoration toward critical connections. Greater reliance on local sourcing may also facilitate adjustment following a disruption by providing access to nearby suppliers. These examples suggest

[2]For instance, there has been a roughly fourfold increase in billion-dollar disasters since the 1980s. NOAA National Centers for Environmental Information, U.S. Billion-Dollar Weather and Climate Disasters (2024), `https://www.ncei.noaa.gov/access/billions/`, DOI: 10.25921/stkw-7w73.

[3]Duranton et al. (2004) provide a broad foundation for viewing urban agglomeration as multidimensional, emphasizing distinct mechanisms of sharing, matching, and learning.

Regarding the labor market concentration, Glaeser et al. (2004) and Behrens et al. (2014) emphasize the fundamental role of skilled human capital in city development. Wage premia arising from firm competition in cities attract additional workers, further expanding urban areas and reinforcing agglomeration economies. Duranton et al. (2001) illustrate the dynamics of city-level specialization, showing that innovation tends to emerge in diversified cities, while mass production often relocates to more specialized cities with lower production costs. In this paper, city diversity is measured as the inverse of the Herfindahl index based on sectoral employment shares within each city.

Regarding transportation infrastructure, Duranton et al. (2012) find that a greater initial stock of highways leads to higher long-run employment and faster economic growth at the city level. Similarly, Baum-Snow (2007) shows that the expansion of the interstate highway system contributed substantially to urban decentralization in the United States between 1950 and 1990.

From the perspective of economic geography and trade, Marin et al. (2021) document that export activity is disproportionately concentrated in larger cities, implying greater potential gains from trade at the urban level and reciprocal benefits for city development.

Furthermore, three agglomeration forces are also closely interconnected. For example, Marin et al. (2021) show that active trade shifts labor supply toward larger cities, while Duranton et al. (2014) find that interstate highways increase the overall volume of exports across U.S. cities, though they have little effect on total export value.

that the economic consequences of natural disasters may vary with the particular characteristics of each dimension of urban economic structure.

This paper examines whether differences in industrial composition, road networks, and trade linkages amplify or mitigate the economic losses caused by natural disasters. In doing so, it moves beyond evaluating urban economic organization solely through the lens of productivity under ordinary conditions to consider resilience as another dimension of urban economic performance. Ultimately, the analysis asks when the structures underlying agglomeration remain an asset in the face of disruption and when they instead become a source of vulnerability. Identifying the heterogeneous ways in which different dimensions of urban economic structure shape disaster impacts requires a carefully designed empirical strategy, considering for the inherently unpredictable and transient nature of natural disasters. To address this challenge, I employ the local projection difference-in-differences (LP-DiD) framework developed by Dube et al. (2023) and Jordà et al. (2025) using a monthly panel of U.S. MSAs spanning 2018 to 2024. The LP-DiD framework accommodates the temporary and recurring nature of natural disasters, allowing cities to experience multiple events over time while constructing appropriate control cities that remain unexposed within each event window.

Measuring disruptions to local economic activity at the monthly frequency and MSA level is challenging because broad measures of aggregate economic activity, such as GDP, are not available at this spatial and temporal resolution. I therefore use monthly commercial and industrial electricity consumption as the primary outcome. As an essential input across business and production activities, electricity consumption captures activity across a wide range of sectors and responds directly to disruptions in business operations[4]. I complement the baseline analysis with monthly employment changes, which capture a distinct margin of economic adjustment through the labor market and provide corroborating evidence for the patterns identified using electricity consumption. Having defined the outcomes used to capture local economic activity, I identify disaster exposure using physical disaster intensity rather than realized monetary losses, which may themselves reflect the economic consequences of interest. Specifically, I define treatment using a threshold for physical disaster intensity benchmarked to Category 2 hurricane wind speeds, thereby restricting treatment to sufficiently severe natural disaster exposure.

This paper yields three main findings. First, natural disasters lead to substantial short-run economic declines. For an MSA in which no one structural feature is particularly pronounced, disaster

[4]Cicala (2023) similarly uses electricity consumption to measure economic activity during the COVID-19 pandemic.

exposure reduces monthly commercial and industrial electricity consumption by approximately 3 percent relative to pre-disaster levels. Valuing the associated loss in electricity service, the median estimated economic loss is equivalent to approximately 14 percent of monthly MSA GDP. Consistent with the decline in electricity consumption, natural disaster exposure reduces three-month average monthly employment growth by approximately 576 workers, equivalent to roughly 54 percent of the average pre-disaster employment growth among treated MSAs.

Second, the economic effects of natural disasters vary substantially with urban structure. Cities with more concentrated labor markets experience larger economic disruptions, whereas those with more centralized road networks and stronger intra-city goods-sourcing capacity exhibit greater resilience. More specifically, a one-standard-deviation increase in industrial concentration deepens the decline in economic activity from 3 percent to approximately 7.3 percent, making the disruption roughly 2.5 times larger than the baseline effect. This heterogeneity may partly reflect differences in how individual industries respond to natural disasters, an interpretation I examine further by decomposing industrial concentration across broad industry groups. In contrast, a one-standard-deviation increase in road network centralization reduces the post-disaster decline in economic activity by approximately 60 percent. Supporting evidence from traffic conditions indicates that more centralized road networks experience smaller increases in vehicle-hours delay during weather-related events, suggesting that their relative effectiveness during disruptions may differ from that under everyday traffic conditions. Similarly, a one-standard-deviation increase in intra-city goods-sourcing capacity reduces the post-disaster decline in economic activity by approximately 44 percent. Cities that source a larger share of goods locally are therefore less exposed to disruptions in goods-sourcing relationships extending beyond the MSA. Taken together, these findings show that distinct dimensions of urban structure can amplify or mitigate the short-run economic consequences of natural disasters.

Finally, I examine what drives the pronounced heterogeneity associated with industrial concentration. Because industries differ in their exposure and ability to adjust to disaster-related disruptions, the resilience consequences of a concentrated labor market may depend on the industries in which that concentration occurs. I therefore decompose overall industrial concentration into the contributions of six broad industry groups, including Retail, Utilities, and Construction, Business Services, Health and Education, Manufacturing, Accommodation, Wholesale Trade, and Transportation, and Others. The results show that Manufacturing, Accommodation, Wholesale Trade, and Transportation, and Health and Education have distinct implications for disaster resilience. Greater concentration in Manufacturing mitigates the economic impacts of natural disasters, whereas greater concentration in Accommodation, Wholesale Trade, and Transportation and in Health and Education exacerbates them. The greater vulnerability of Accommodation, Wholesale Trade, and Transportation group is consistent with its prominent role as a destination in individuals' daily mobility patterns, such that disruptions to these activities may have broader implications for routine economic activity. For Health and Education, healthcare establishments face substantially higher economic costs from a given electricity interruption than other non-residential establishments, providing one potential explanation for the greater vulnerability. These findings show that the resilience consequences of industrial concentration depend not only on its degree but also on where that concentration occurs. Taken together, the findings of this paper suggest that the benefits of agglomeration cannot be evaluated solely through productivity, as urban economic structures that perform well under normal conditions may differ substantially in their vulnerability to natural disasters.

This paper contributes to the literature on urban economic structure and disaster resilience along four dimensions. First, the industrial-composition results contribute to the long-standing debate over specialization and diversification in cities, which emphasizes competing benefits from within-industry specialization and cross-industry diversity (Marshall, 1890; Jacobs, 1969; Glaeser et al., 1992). Duranton et al. (2001) show that diversified cities facilitate experimentation and innovation, while specialized cities become advantageous once production processes mature. More recently, Heblich et al. (2025) document a dynamic tradeoff in which industrial concentration can generate short-run gains while diversification improves longer-run urban performance. This tradeoff further extends to resilience against natural disasters. Coulson et al. (2020) show that economically diversified metropolitan areas experience smaller and shorter-lived disaster effects. Building on this evidence, my results provide further support for diversification over specialization specifically in shaping resilience to natural disasters, while showing that the consequences of industrial concentration within this setting depend importantly on the industries in which that concentration occurs.

Second, the road-network results contribute to the literature on transportation infrastructure and urban economic activity. While the importance of transportation infrastructure for urban growth is well established (Duranton et al., 2012; Tyndall, 2021), recent work has shifted attention from the presence of infrastructure to its configuration within transportation networks. More specifically, the allocation and economic importance of individual network links (Fajgelbaum et al., 2020; Allen et al., 2022) and their interaction with individual mobility (Miyauchi et al., 2025) has been highlighted. More closely related to disaster resilience, Bergantino et al. (2024) show that the configuration of road networks shapes their vulnerability to disruptions. I extend this line of research by showing that road-network configuration also has important implications for the resilience of local economic activity to natural disasters. Greater centralization of economically important road connections mitigates disaster-induced economic losses, consistent with actual disaster-response practices that prioritize critical routes in restoring essential traffic (Federal Highway Administration, 2013). As an additional contribution, I construct a measure that combines road-network centrality with workplace dependence on workers' physical presence to capture the economic relevance of road-network structure.

Third, I contribute to the literature on trade linkages under natural disasters by showing that their geographic organization can serve as a source of local economic resilience. Existing work shows that natural-disaster shocks propagate through production networks (Caliendo et al., 2018; Carvalho et al., 2021), while Todo et al. (2015) show that supply-chain relationships can also facilitate recovery following the Great East Japan Earthquake. The geography of these relationships can further shape adjustment. Miyauchi (2024) shows that firms rematch with alternative suppliers more quickly where potential suppliers are geographically denser, while Kawakubo et al. (2024) document increased formation of nearby supplier relationships following supply-chain disruptions from the Great East Japan Earthquake. Building on these findings, I show that greater reliance on intra-city sourcing mitigates disaster-induced economic losses.

Finally, I contribute to the literature on industry-specific sources of urban growth by showing that industries also differ systematically in their resilience to natural disasters. Existing work has documented substantial heterogeneity across industries in the benefits they derive from and contribute to urban agglomeration, with particular attention to the growing importance of cognitive and non-routine-intensive activities in shaping urban growth and productivity (Monte et al., 2023; Eckert et al., 2022; Hsieh et al., 2023; Jarosch et al., 2021; Rossi-Hansberg et al., 2021; Fajgelbaum et al., 2020; Rossi-Hansberg et al., 2019). I extend this literature by documenting industry-specific differences in disaster resilience. Greater concentration in Manufacturing mitigates disaster-induced economic losses, whereas concentration in Accommodation, Wholesale Trade, and Transportation and in Health and Education amplifies them.

The remainder of the paper is organized as follows. Section II develops the conceptual framework for multidimensional urban economic structure. Section III describes the data and construction of the main measures. Section IV presents the empirical setting and LP-DiD research design. Section V reports the main results on urban structure and disaster resilience. Section VI examines heterogeneity across industry groups. Sections VII and VIII provide supporting evidence from individual mobility patterns and road-network performance, respectively. Section IX translates the estimated reductions in non-residential electricity consumption into economic losses relative to local GDP. Section X concludes.

## II Conceptual Framework

The economics of agglomeration has long emphasized that the organization of economic activity within cities cannot be reduced to a single dimension. Duranton et al. (2004), for example, distinguish among multiple microeconomic mechanisms underlying agglomeration, including input sharing, labor-market matching, and knowledge transmission. More recent spatial models extend this perspective by explicitly characterizing economic activity through interactions in factor and goods markets and through the spatial linkages connecting workers, firms, and locations (Redding et al., 2017). Transportation and commuting constitute an important component of these linkages within cities. In Lucas et al. (2002), for example, firms balance the agglomeration benefits of locating near other producers against the commuting costs faced by workers, making worker accessibility an integral component of the internal organization of economic activity.

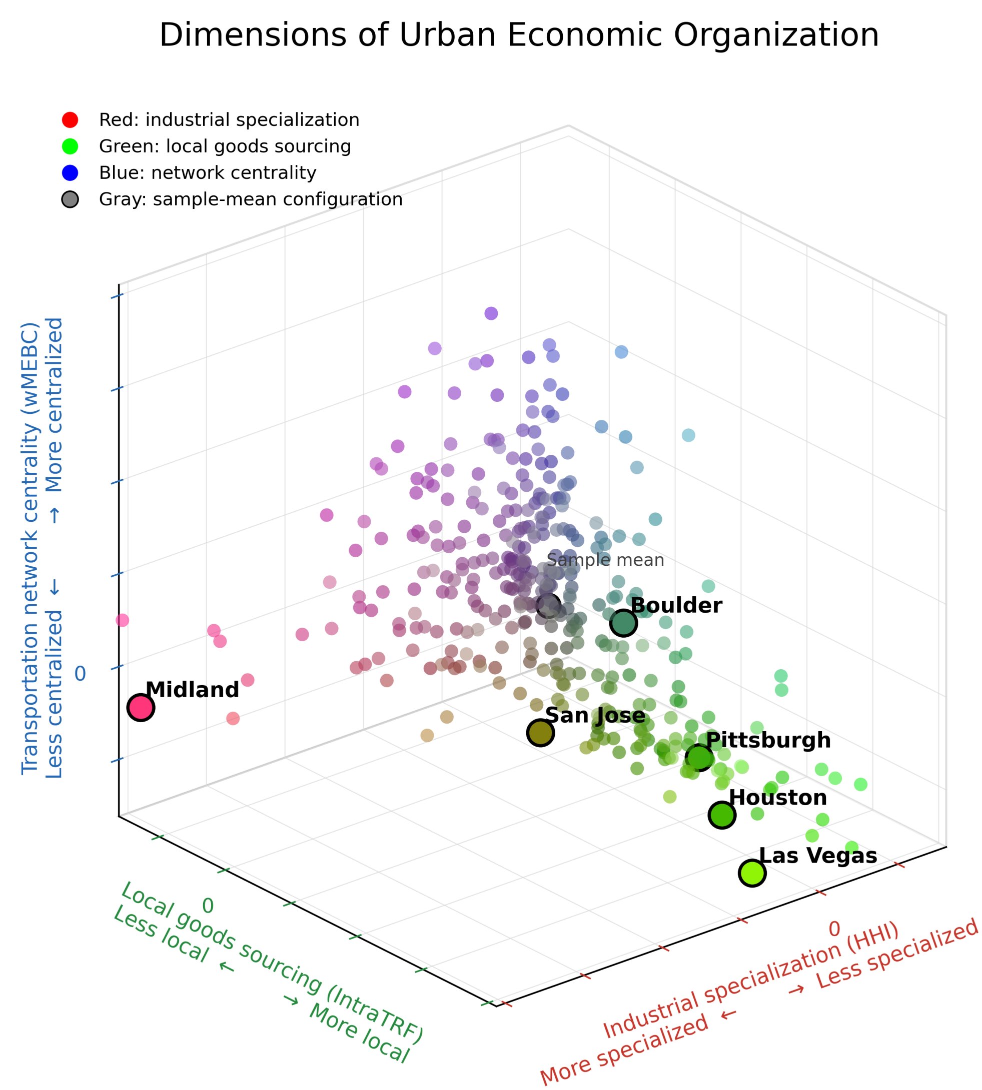
Taken together, this literature points to three complementary dimensions of urban economic structure that are central to local production, including industrial composition, the transportation linkages connecting workers to production, and the trade linkages connecting firms to goods and inputs. Because these dimensions need not move together, cities with similar levels of overall spatial concentration may nevertheless exhibit fundamentally different patterns of urban economic structures. Figure 1 illustrates this multidimensionality using the three dimensions considered in this paper: labor-market concentration, road-network centrality, and within-city goods sourcing. The figure reports their standardized values across all 365 MSAs in the sample and highlights several illustrative cases.

Consider the San Jose, California, and Houston, Texas, MSAs. Despite similar levels of overall spatial concentration, the two cities exhibit both similarities and differences in their underlying urban economic structures. Both have relatively dispersed road networks and high shares of within-city goods sourcing, yet they differ sharply in industrial composition. San Jose exhibits a high degree of industrial concentration, with employment clustered in high-technology industries and producer services, including finance and professional services. Houston, by contrast, has a more diversified employment base, with no small group of industries accounting for a dominant share of local employment.

The economic relevance of this multidimensionality extends beyond the organization of activity in normal times. These multiple, interconnected dimensions take on additional significance when shocks occur. They provide distinct margins through which local economies adjust to shocks. For instance, Monte et al. (2018) first jointly incorporate goods and factor markets, commuting, and migration to characterize the multidimensional organization of local economies. They then show how one of these interconnected dimensions can provide a margin of adjustment to local shocks, finding that commuting openness serves as an important adjustment margin following local labor-demand shocks. Similarly, Caliendo et al. (2018) show that regional and sectoral shocks propagate differently depending on the spatial structure of the economy, jointly characterized by interregional and intersectoral trade linkages, sectoral composition, and factor mobility. Their application to the destruction of infrastructure in Louisiana following Hurricane Katrina further illustrates how these interconnected dimensions shape the adjustment of economic activity to a localized shock.

Taken together, local economies are organized simultaneously along multiple dimensions of economic structure. At the same time, these underlying dimensions may themselves provide distinct margins through which local economies adjust to shocks. This paper brings these perspectives together by examining whether pre-existing differences across three dimensions of urban economic structure systematically shape the short-run response of local economic activity to natural disasters.

Figure 1: Cities Differ in How Economic Activity Is Organized




*Notes:* Each point represents a U.S. MSA. The three axes summarize distinct urban economic structures: industrial concentration ($HHI$), road-network structure (weighted edge betweenness centrality; $wMEBC$), and intra-city trade ($IntraTFR$). Because all three measures are standardized, zero on each axis denotes the sample mean of that dimension, while the gray point at the origin represents the joint sample-mean configuration. Point colors provide an additional representation of each MSA's urban economic structures across the three dimensions: red, green, and blue intensities correspond to $HHI$, $IntraTRF$, and $wMEBC$, respectively, with darker (brighter) intensities indicating values further below (above) the sample mean. The resulting color mixture therefore reflects each MSA's multidimensional urban configuration. Selected MSAs illustrate that cities with similar characteristics along one dimension may differ substantially along others, motivating the multidimensional characterization of urban structure used throughout the paper.

# III Data

## III.I Measures of Economic Activity

Measuring local economic activity at a monthly frequency and the MSA level is challenging because conventional measures of aggregate economic activity, such as GDP, are not available at this spatial and temporal resolution. This measurement challenge is particularly important in the context of natural disasters, where disruptions can occur immediately and may dissipate or evolve within a relatively short period. I therefore use monthly commercial and industrial electricity consumption as the primary measure of local economic activity. Electricity is an essential input across a broad range of economic activities, so changes in local economic activity can be reflected relatively quickly in electricity demand[5]. I focus on commercial and industrial electricity consumption to more directly capture economic activity associated with local businesses and production, excluding residential and transportation consumption.

Using electricity consumption to measure economic activity nevertheless introduces an important measurement concern in the context of natural disasters. Observed electricity consumption may change not only because underlying economic activity changes, but also because disasters damage the electricity system. Electricity demand also responds directly to weather conditions, particularly through heating and cooling needs. I therefore account explicitly for electricity outages and climatological conditions in constructing the empirical specifications below. These controls help distinguish changes in electricity consumption associated with local economic activity from variation arising directly from electricity-system disruptions and weather.

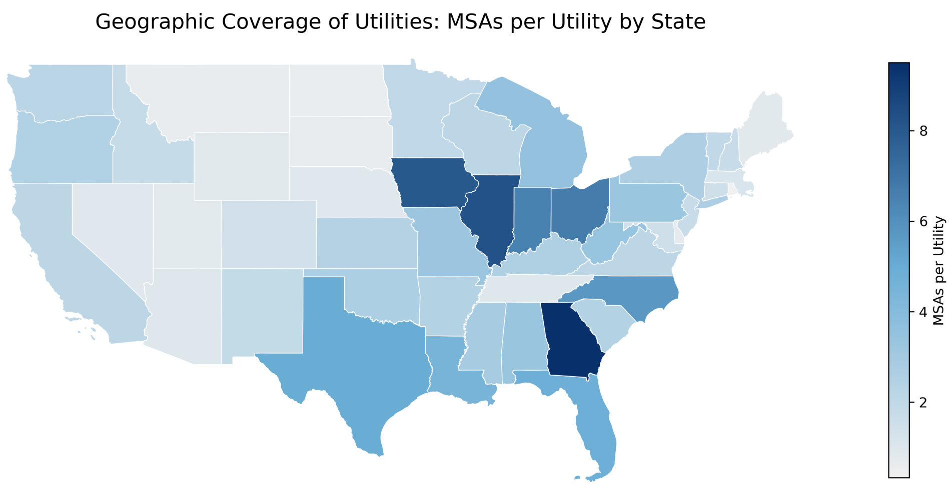
Employment provides a more conventional measure of local economic conditions that is also available at the monthly frequency and MSA level. However, employment reflects adjustment specifically through the labor market rather than changes in economic activity more broadly. This distinction is particularly relevant for measuring the immediate effects of natural disasters. Firms may experience substantial disruptions to their operations without immediately adjusting their workforce. Employment may therefore respond more gradually to short-run disruptions than electricity consumption. For this reason, I use commercial and industrial electricity consumption as the primary measure of local economic activity and complement the main analysis with employment-based outcomes to examine whether the estimated effects across urban economic structures are also reflected in local

[5]Cicala (2023) demonstrates that electricity consumption across industrial, commercial, transportation, and residential sectors closely tracks economic activity, particularly during the COVID-19 pandemic.

labor market adjustment.

I use the U.S. Energy Information Administration's (EIA) Form 861M, the Monthly Electric Power Industry Report, and Form 861, the Annual Electric Power Industry Report. Electricity sales data are available at the utility-month level while a utility's service territory frequently spans multiple counties and may extend across multiple MSAs and state boundaries. Consequently, utility service territories do not generally align with MSA boundaries, requiring the construction of MSA-level measures from utility-level electricity sales data. Figure 2 presents the geographic scope of utilities varies substantially across states. On average, a single utility within one state serves approximately 2.7 MSAs, with a median of 2.1 and a range from 0.3 to 9.5 MSAs.

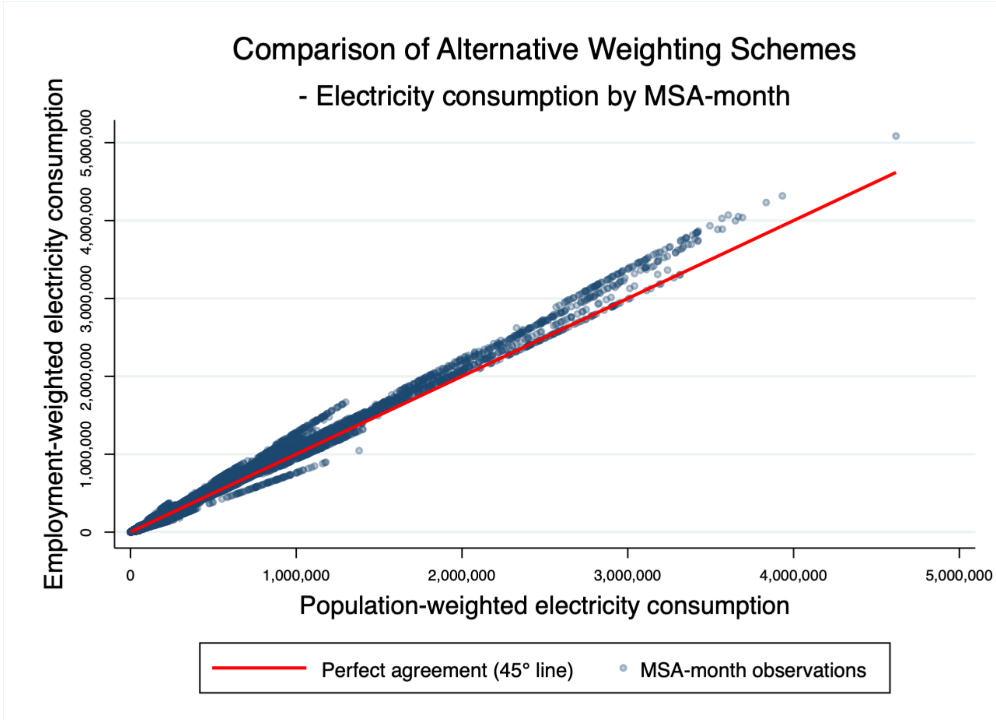
Figure 2: Geographic Coverage of Utilities: MSAs per Utility by State




*Note*: Darker shades indicate that a single utility serves a larger number of MSAs within a state.

To construct MSA-level electricity consumption measures, I combine utility-level sales data with the Service Territory Information schedule from EIA Form 861, which identifies the counties served by each utility. I then allocate utility-level electricity sales across counties using county population weights and aggregate the resulting county-level estimates to the MSA level. As a robustness check, I also construct MSA-level electricity consumption using county employment shares from the County Business Patterns (CBP) as an alternative weighting scheme. Figure 3 compares electricity consumption measures constructed using population and employment weights. The two measures exhibit a close one-to-one relationship, indicating that the electricity consumption measure is largely robust to the choice of weighting scheme.

Figure 3: Comparison of Electricity Consumption Measures under Alternative Weighting Schemes




*Note*: Each point represents an MSA-month observation. The red 45° line indicates perfect agreement between the two weighting schemes. The close alignment of observations with the 45° line suggests that the electricity consumption measure is largely insensitive to whether population or employment weights are used.

### III.I.1 Additional Controls for Electricity Consumption

As discussed above, observed electricity consumption may respond directly to electricity-system disruptions and climatological conditions in addition to changes in underlying economic activity by natural disasters. I therefore construct MSA-month-level measures of electricity outages and weather conditions for inclusion in the empirical specifications.

To measure electricity-system disruptions, I use outage data from the Environment for the Analysis of Geo-Located Energy Information (EAGLE-I) program developed by Oak Ridge National Laboratory (ORNL) for the U.S. Department of Energy (DOE). The dataset provides county-level outage information from 2014 to 2025, recording the number of customers without power at each observation. I construct an outage-intensity measure by counting the number of recorded outage observations within each county-month and dividing by the number of calendar days in that month. I then aggregate this measure to the MSA-month level by taking the mean across counties within each MSA. The resulting measure captures the average number of recorded outage observations per day across counties within an MSA. To account for variation in electricity demand associated with weather, I additionally construct MSA-month-level heating and cooling degree days using data from the National Oceanic and Atmospheric Administration (NOAA).

Finally, I account for differences in the local building stock using the MSA-month average share of buildings constructed before 1980 from the U.S. Census Bureau's American Community Survey (ACS). Building vintage may affect electricity demand through differences in insulation, air conditioning, energy efficiency, and other building characteristics. I therefore include the pre-1980 building share as an additional control for differences in electricity demand associated with the local building stock. Building vintage may also contribute to heterogeneity in disaster resilience, both through differences in the physical vulnerability of older and newer structures and through broader differences in urban development associated with building vintage. I examine these possibilities separately in Appendix XI.II.

#### III.I.2 Employment

To examine whether disaster-induced disruptions are also reflected in local labor market adjustment, I complement the electricity-based measure of economic activity with employment outcomes. Specifically, I use MSA-month-level three-month average employment growth from the Current Employment Statistics–State and Area Employment (CES-SAE) program[6], although labor market outcomes capture only labor adjustment following natural disasters and are therefore narrower than electricity consumption as a measure of aggregate economic activity. For measuring short-run labor market responses to natural disasters, changes in employment are more informative than employment levels because they capture adjustments occurring around the time of the disruption. This emphasis is consistent with Gentili (2025), who measure the immediate economic impacts of the May 2023 floods in Italy using changes in high-frequency indicators of economic activity of new daily hires.

### III.II Natural Disaster Intensity

I use observed wind speeds as a physical measure for identifying high-intensity natural-disaster exposure. Compared with monetary losses, which are mechanically influenced by local economic activity and the value of exposed assets, wind speed provides a more objective measure of the underlying physical intensity of a disaster. The use of wind intensity as an objective physical measure

[6] The QCEW provides near-universal coverage of employment and wages based on unemployment insurance records, making it a comprehensive measure of local labor market conditions. In contrast, the CES-SAE program offers more timely indicators of labor market dynamics, particularly for nonfarm employment.

of disaster exposure is well-established in the literature, particularly in studies of hurricanes, where locally experienced wind speeds are commonly used to characterize the severity of exposure. Wind intensity can also capture dimensions of disaster severity beyond direct wind accompanying disasters, as stronger winds can amplify other hazards, such as storm surge and flooding, and contribute to the spread of wildfires(Botzen et al. (2019), Strobl (2011)).

The primary data sources for wind speed records are the Storm Events Database maintained by NOAA's National Centers for Environmental Information (NCEI) and the Atlantic Hurricane Database (HURDAT2) provided by NOAA's National Hurricane Center (NHC). I combine the wind observations from these two datasets and assign them to MSAs based on their locations. I then construct MSA-month disaster intensity by averaging the wind intensities of all recorded event-level observations within an MSA during that month. Thus, the resulting measure captures the average event-level wind intensity within each MSA-month[7].

Using the average event-level wind intensity constructed for each MSA-month, I identify high-intensity natural-disaster treatment status based on a common physical intensity benchmark. Specifically, an MSA-month is classified as being treated when its average event-level wind intensity exceeds 96 mph, corresponding to the lower-bound wind intensity associated with Category 2 on the Saffir–Simpson scale. The 96 mph threshold is therefore used as a benchmark for classifying the intensity of the constructed MSA-month exposure measure, rather than to classify each underlying wind observation as a Category 2 hurricane. Table 1 provides descriptive context for this threshold by reporting the monthly frequency with which MSA-month wind intensity reaches different wind-intensity benchmarks corresponding to the Saffir–Simpson scale between 2018 and 2024. The frequency declines sharply as the wind-intensity benchmark increases, while exposure at the threshold used in the main analysis remains distributed throughout the year.

Additionally, I use average event-level wind intensity instead of the maximum wind intensity recorded within an MSA-month. A maximum-based measure can be particularly problematic with monthly economic outcomes when the strongest event occurs near the end of the month. Although relatively little of that month's economic activity is exposed to the event, its wind intensity would nevertheless determine the monthly maximum. Under the average measure, by contrast, such an

[7]In the Storm Events Database, recorded wind speeds are available for event types including High Wind, Marine High Wind, Marine Strong Wind, Marine Thunderstorm Wind, Strong Wind, and Thunderstorm Wind, which are classified separately from tropical storms, hurricanes, tornadoes, floods, and hail. I exclude tornado events because direct wind speed measurements are not available. Tornado intensity is instead classified using the Enhanced Fujita (EF) scale, which assigns wind-speed ranges based on observed post-event damage. Because these wind speeds are inferred from damage rather than directly observed, tornado exposures are excluded from the analysis. For Atlantic hurricanes, HURDAT2 provides sustained maximum wind speeds, which I use to measure hurricane wind intensity.

event contributes to the MSA-month intensity together with the other recorded events rather than determining the monthly measure on its own.

Table 1: Monthly Frequency of Natural Disaster Exposure by Wind Intensity (2018–2024)

| Month | Intensity 1 | Intensity 2 | Intensity 3 | Intensity 4 | Intensity 5 |
|---|---|---|---|---|---|
| Jan | 54 | 6 | 1 | | |
| Feb | 36 | 2 | 2 | | |
| Mar | 44 | 2 | 1 | | |
| Apr | 43 | 2 | | | |
| May | 25 | 1 | | | |
| Jun | 29 | 1 | | | |
| Jul | 27 | 1 | | 1 | |
| Aug | 30 | 5 | 1 | 1 | |
| Sep | 23 | 3 | | 1 | |
| Oct | 26 | 2 | 1 | 1 | 1 |
| Nov | 29 | 1 | | | |
| Dec | 46 | 2 | 1 | | |
| Total | 412 | 28 | 7 | 4 | 1 |

*Notes:* Intensity levels correspond to the wind-speed ranges of Categories 1–5 on the Saffir–Simpson hurricane wind scale.

### III.III Urban Economic Structures

I construct three dimensions of urban economic structure from pre-period measures calculated using 2017 data. First, to capture the concentration of employment across industries within a city, I calculate the Herfindahl–Hirschman Index ($HHI$) of industrial composition. This measure follows established approaches to characterizing regional industrial concentration and diversification in local labor markets, with Coulson et al. (2020) further applying the complement of the $HHI$ to measure industrial diversification in the context of natural disasters. Second, to capture road-network centrality after accounting for observed commuting patterns, I compute weighted Mean Edge Betweenness Centrality ($wMEBC$). Betweenness centrality is widely used to characterize the structural importance of links within transportation networks and has been applied specifically to assess road-network robustness and resilience to disruptions (Zhang et al., 2015; Tachaudomdach et al., 2021). Building on this established network measure, I develop a commuting-weighted measure of road-network centrality that incorporates observed commuting patterns to capture the road links most relevant to local worker mobility, consistent with the importance of worker accessibility discussed in Section II. Finally, to measure the extent to which a city relies on internal sourcing of goods, I construct the Intra-city Trade Flow Ratio ($IntraTFR$). This measure builds on the use of spatial sourcing shares to characterize the geographic structure of supply relationships (Castro-Vincenzi et al., 2024), with $IntraTFR$ capturing the share of goods entering an MSA that are sourced from within the same MSA.

### Labor Market Job Concentration

Regarding industrial composition, the key question is how concentrated employment is across industries within a city. The Herfindahl-Hirschman Index ($HHI$) of the labor market measures the degree of employment concentration across industries within a city. MSA-month-industry-wise job counts are from the Quarterly Census of Employment and Wages (QCEW), which provides industry-specific employment data classified under the North American Industry Classification System (NAICS). Offered job counts are summarized at the NAICS code 4-digit, industry group level. As a result, the $HHI$ of the labor market is defined as follows:

$$
\begin{aligned}
HHI_{c,m,y} &= \sum_{i=1}^{N} s_{i,c,m,y}^2 \\
&= \sum_{i=1}^{N} \left(\frac{J_{i,c,m,y}}{J_{c,m,y}}\right)^2
\end{aligned}
\tag{1}
$$

where $N$ represents the total number of industries in city $c$, $s_{i,c}^2$ represents the employment share of industry $i$ in city $c$, month $m$ and year $y$. To be more specific, $J_{i,c,m,y}$ represents the number of jobs in industry $i$ and $J_{c,m,y}$ represents the total number of jobs across all industries. The 2017 average $HHI$, which ranges from 0 to 1, measures the degree of industry concentration, with higher values indicating greater industrial concentration and lower employment diversity. For example, the Midland MSA in Texas has the second-highest average HHI among all MSAs, reflecting a highly concentrated labor market with limited industry diversification. This pattern is driven by the concentration of employment in the oil and gas sector and related industries, which together account for more than 50 percent of the local workforce[8].

### Road Network Concentration

For the road network, a representative measure widely used in the transportation (**?**Marshall, 2016) and social-network (Qian et al., 2017) literature is Edge Betweenness Centrality ($EBC$), which

[8] Midland MSA is home to major oil companies, including EOG Resources, Concho Resources, Pioneer Natural Resources, Chevron, Shell, ExxonMobil, and Occidental Petroleum.

captures the structural importance of individual links within a network. More specifically, $EBC$ measures the extent to which a road segment lies on the shortest paths between pairs of locations. Road segments with higher $EBC$ are traversed by a larger share of shortest paths and therefore play a more central role in connecting locations throughout the network.

Following this conventional network representation, I model road segments as edges and intersections as nodes and evaluate the importance of each road segment based on its role in connecting locations throughout the metropolitan network. However, the physical structure of the road network alone does not capture the economic importance of accessibility across locations. Recent literature highlights the role of commuting patterns in shaping localized economic outcomes (Monte et al., 2018)[9]. This distinction is particularly relevant for workers in occupations that require physical presence at the workplace, who rely more directly on transportation access than workers in occupations that can more readily accommodate remote work. Building on the conventional $EBC$ measure, I therefore develop a commuting-adjusted measure of road-network centrality that gives greater weight to network centrality associated with workplace census blocks characterized by greater dependence on on-site work within each MSA.

Census-block-level commuting patterns are drawn from the LEHD Origin-Destination Employment Statistics (LODES), which provide home-workplace block pairs and classify workers into three broad industry groups: (1) goods-producing industries, (2) trade, transportation, and utilities, and (3) all other services. I use the workplace block as the relevant location within each MSA and observe the industry composition of workers commuting to each block. This information allows me to identify workplace locations where employment is relatively more dependent on physical presence. Specifically, I classify workers in goods-producing industries and trade, transportation, and utilities as relatively more constrained to on-site work[10].

I combine this workplace-level commuting information with the physical road network in three steps. First, I calculate $EBC$ for individual road segments using road networks obtained from OpenStreetMap (OSM) through the OSMnx module in Python (Boeing, 2017a). Second, I spatially assign these edge-level centrality values to workplace census blocks to characterize the road-network centrality associated with each workplace location. Third, I give greater weight to workplace blocks

[9]Monte et al. (2018) show that variation in commuting patterns serves as an adjustment margin through which the localized effects of shocks are redistributed across space. In particular, heterogeneous commuting patterns across counties play a key role in shaping local employment responses to labor-demand shocks.

[10]The remaining category includes producer services such as finance and insurance that are generally less constrained to on-site work. This classification is necessarily imperfect because it also contains physically constrained sectors such as accommodation, health, and education. I therefore interpret the classification as capturing relative differences in physical-presence requirements across the three industry groups available in LODES.

with larger shares of workers in sectors requiring physical presence and aggregate the resulting measure to the MSA level.

Let $N_c$ denote the complete road network of MSA $c$, $e$ a road segment in that network, and $b$ a workplace census block, with $\mathcal{B}_c$ denoting the set of workplace blocks in MSA $c$. I first calculate $EBC(e; N_c)$ for each road segment based on its position within the complete metropolitan road network. I then map this edge-level measure to workplace blocks. Let $E_b(N_c) \subseteq E(N_c)$ denote the set of road segments spatially associated with workplace block $b$. The road-network centrality associated with block $b$ is defined as the average $EBC$ of these road segments:

$$EBC_b(N_c) = \frac{1}{|E_b(N_c)|} \sum_{e \in E_b(N_c)} EBC(e; N_c) \tag{2}$$

Thus, $EBC_b(N_c)$ assigns the centrality of the underlying metropolitan road network to a workplace location rather than calculating centrality over census blocks themselves.

Next, I measure the physical-presence intensity of employment at each workplace block. For each block $b$, I calculate the share of workers employed in sectors with relatively high physical-presence requirements.

$$\text{Share}_b^{HP} = \frac{\sum_{s \in \mathcal{S}^{HP}} J_{b,s}}{\sum_{s \in \mathcal{S}} J_{b,s}} \tag{3}$$

Here, $J_{b,s}$ denotes the number of workers employed in industry group $s$ at workplace block $b$, obtained by aggregating workers commuting to block $b$ from all origin blocks. Formally,

$$J_{b,s} = \sum_{o} OD_{o \to b,s} \tag{4}$$

where $OD_{o \to b,s}$ denotes the number of workers commuting from origin block $o$ to workplace block $b$ in industry group $s$. $\mathcal{S}$ denotes the set of all industry groups, and $\mathcal{S}^{HP} \subseteq \mathcal{S}$ denotes the subset with relatively high physical-presence requirements.

I then combine the road-network and workplace-employment components at the block level:

$$wEBC_b = EBC_b(N_c) \cdot \text{Share}_b^{HP} \tag{5}$$

A workplace block therefore receives a larger $wEBC_b$ when it is both associated with more central road segments and characterized by a larger share of workers whose jobs require physical presence.

Finally, I aggregate across workplace blocks to construct the MSA-level commuting-adjusted road-network centrality measure:

$$wMEBC_c = \frac{\sum_{b \in \mathcal{B}_c} EBC_b(N_c) \cdot \text{Share}_b^{HP}}{\sum_{b \in \mathcal{B}_c} \text{Share}_b^{HP}}. \tag{6}$$

The resulting $wMEBC_c$ is a time-invariant MSA-level measure of road-network centrality that places greater weight on workplace locations with greater dependence on physically present workers. Higher values indicate that an MSA's on-site-dependent workplace locations are, on average, associated with more structurally central parts of the metropolitan road network. Because such segments play a disproportionate role in connecting locations throughout the network, their disruption from traffic accidents, road construction, or severe weather can have broader implications for urban accessibility (Mattsson et al., 2015).

Figure 4 presents Flagstaff, Arizona, which has the highest $wMEBC$ among the MSAs in the sample. The left panel shows that Flagstaff's road-network centrality is concentrated along a limited set of major corridors. In particular, relatively high $EBC_b$ values trace the north–south and east–west corridors that converge in the south-central portion of the MSA, with additional high-centrality segments extending northward. The middle panel shows that workplace dependence on physical presence is considerably more localized. Several workplace blocks with high physical-presence shares are located near the south-central intersection of the major corridors, while other high-share workplace locations are distributed elsewhere in the MSA.

The right panel shows how these two spatial patterns combine. Because the physical-presence share is bounded between zero and one[11], the weighting does not amplify the underlying road-network centrality. Rather, it selectively attenuates $EBC_b$ according to the degree of workplace dependence on physical presence. Locations therefore remain prominent in the right panel when relatively high road-network centrality coincides with a relatively high physical-presence share. This pattern is particularly visible around the south-central intersection and along portions of the major corridors leading into it. By contrast, some high-centrality locations visible in the left panel become substantially attenuated when their associated workplace blocks have relatively low physical-

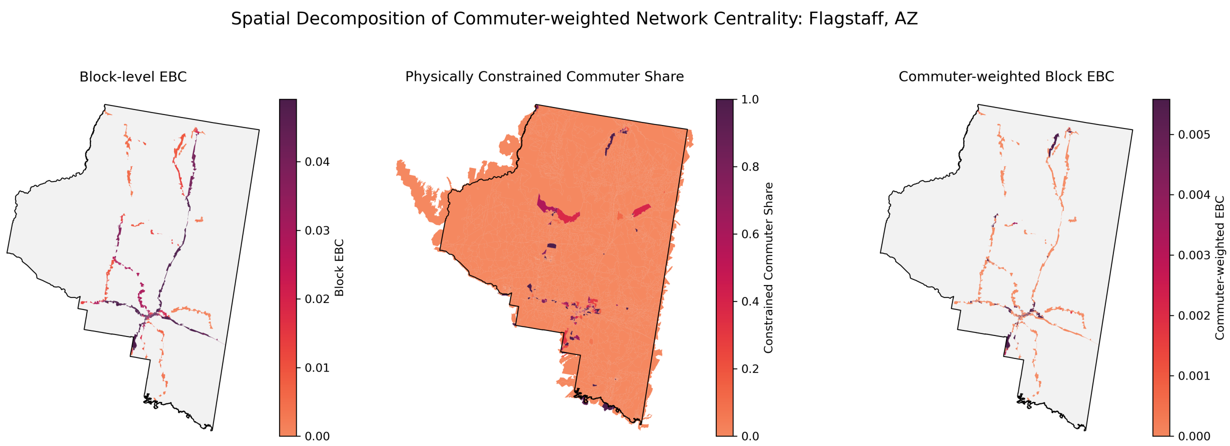
[11] More precisely, because $0 \leq \text{Share}_b^{HP} \leq 1$, the block-level weighted centrality satisfies

$$wEBC_b = EBC_b \times \text{Share}_b^{HP} \leq EBC_b. \tag{7}$$

Thus, the physical-presence weighting can only attenuate, rather than amplify, the underlying block-level road-network centrality.

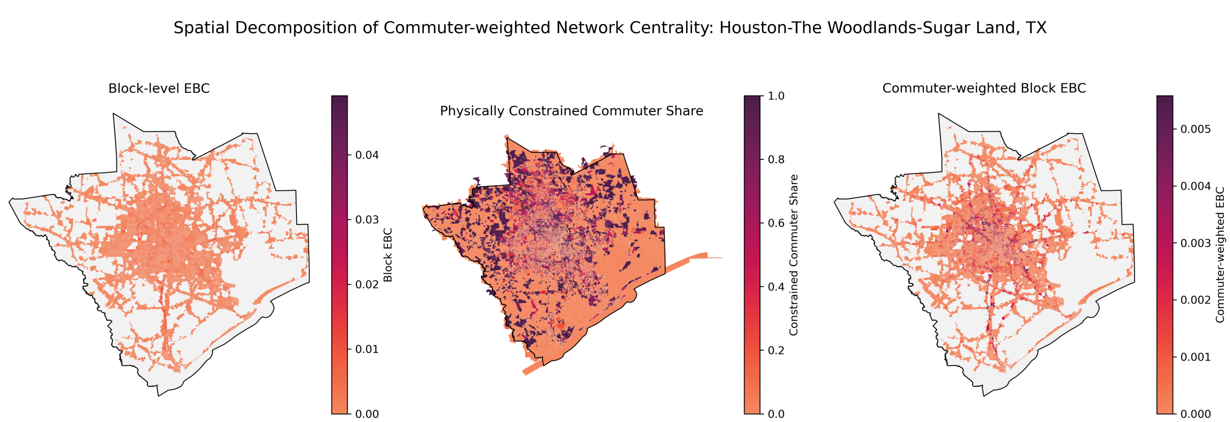
presence shares. Similarly, workplace blocks with high physical-presence shares do not receive high $wEBC_b$ when they are associated with relatively low-centrality portions of the road network. Flagstaff's high MSA-level $wMEBC$ therefore reflects the spatial coincidence of a centralized road-network structure with workplace locations that are relatively dependent on on-site employment.

Figure 4: Highest $wMEBC$: Flagstaff, AZ




In contrast, Figure 5 presents Houston–The Woodlands–Sugar Land, Texas, which has the lowest $wMEBC$ among the MSAs in the sample. Unlike Flagstaff, block-level road-network centrality is distributed across a dense and spatially extensive network rather than concentrated along a small number of dominant corridors. Workplace locations with relatively high physical-presence shares are similarly dispersed across the metropolitan area. Because both components are spatially diffuse, their combination in the right panel also remains broadly distributed rather than concentrated around a limited set of central locations. Consequently, Houston exhibits a substantially lower MSA-level $wMEBC$.

Figure 5: Lowest $wMEBC$: Houston-Sugar Land-Baytown, TX



### Intra-city Goods Trade Flow Concentration

Lastly, the flow of supplied goods plays a pivotal role in sustaining a city's economic activity. I construct a measure of goods flow concentration, the intra-city traded flow ratio ($IntraTFR$), which captures the relative importance of internally sourced trade linkages. Specifically, $IntraTFR$ reflects the extent to which goods delivered to city $c$ are sourced from within the same city, relative to all metro-origin shipments delivered to that city. The Freight Analysis Framework Version 5 (FAF5) Experimental County-Level Estimates, provided by the Bureau of Transportation Statistics, offer freight flow information by transportation mode and commodity group. While FAF5 reports shipment estimates at a coarser geographic level, an experimental county-level disaggregation factor, available only for 2022, is used to approximate county-to-county truck flows. To construct the pre-period $IntraTFR$ measure, I assume that the 2022 county-level disaggregation factors are unchanged from 2017. Applying this disaggregation factor, I first construct county-pair shipment volumes and then aggregate them to the MSA-pair level. In this step, I take the average shipment volume across all county pairs within each origin-MSA, destination-MSA, and commodity-group cell, capturing the average intensity of trade linkages between MSAs. I then aggregate these MSA-pair trade intensities across commodity groups to construct a city-level measure.

Based on this, the Intra-city Traded Flow Ratio ($IntraTFR$) is defined as follows:

$$IntraTFR_{c,y} = \frac{\sum_{j=c} T_{j \to c,y}}{\sum_j T_{j \to c,y}} \tag{8}$$

where $T_{j \to c,y}$ denotes the MSA-to-MSA shipment intensity from origin MSA $j$ to destination MSA $c$ in year $y$, constructed from average county-to-county shipment volumes within each MSA pair and aggregated across commodity groups. Accordingly, $\sum_{j=c} T_{j \to c,y}$ captures the inward trade intensity sourced from within MSA $c$, while $\sum_j T_{j \to c,y}$ captures the total inward trade intensity from all metro origins. Table 2 summarizes the scope of each urban economic structure and its economic implications.

## III.IV Correlation Across Three Dimensions of Urban Economic Structure

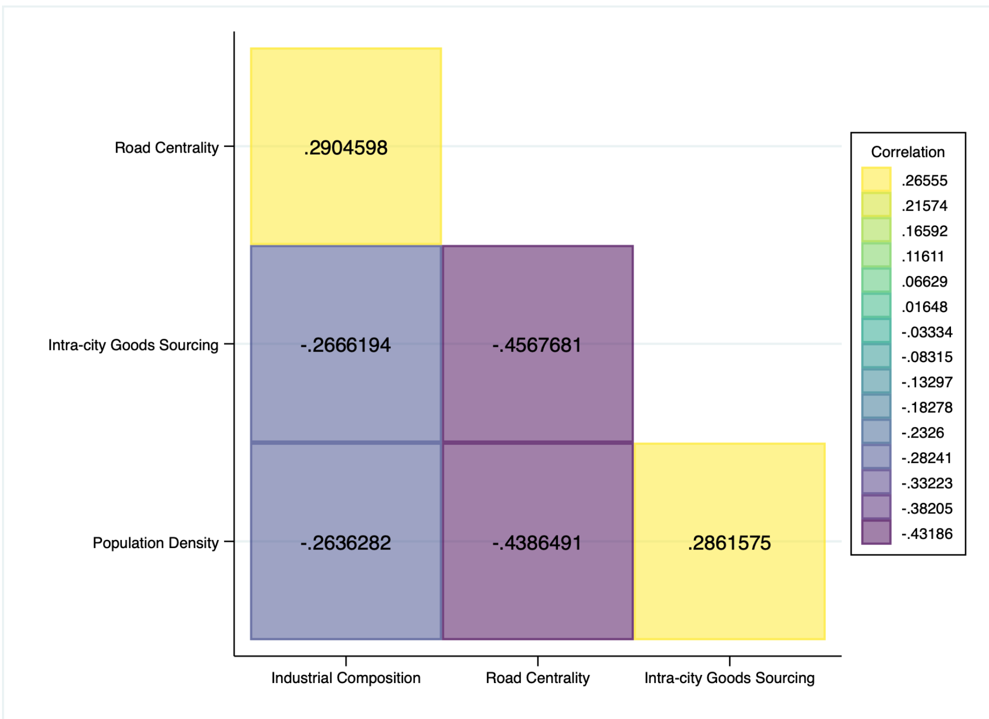
The three dimensions of urban economic structure, including industrial composition, road network structure, and intra-city trade linkages, capture different margins of urban economic organization beyond the overall agglomeration intensity reflected in population density. Figure 6 presents the correlations among the corresponding measures, $HHI$, $wMEBC$, and $IntraTFR$. The measures

Table 2: Interpretation of Agglomeration Forces

| Measure | Range | Interpretation (higher values) |
|---|---|---|
| Labor Market ($HHI$) | $[0, 1]$ | Fewer industries account for most jobs in the city |
| Road Network ($wMEBC$) | $[0, 1]$ | A few key road segments carry most connections across locations, especially for on-site workers |
| Trade Flow ($IntraTFR$) | $[0, 1]$ | More goods are sourced within the city |

do not exhibit a high degree of overlap, consistent with each capturing a distinct feature of how economic activity is organized within an MSA.

Figure 6: Correlation Matrix of Agglomeration Measures at the MSA Level

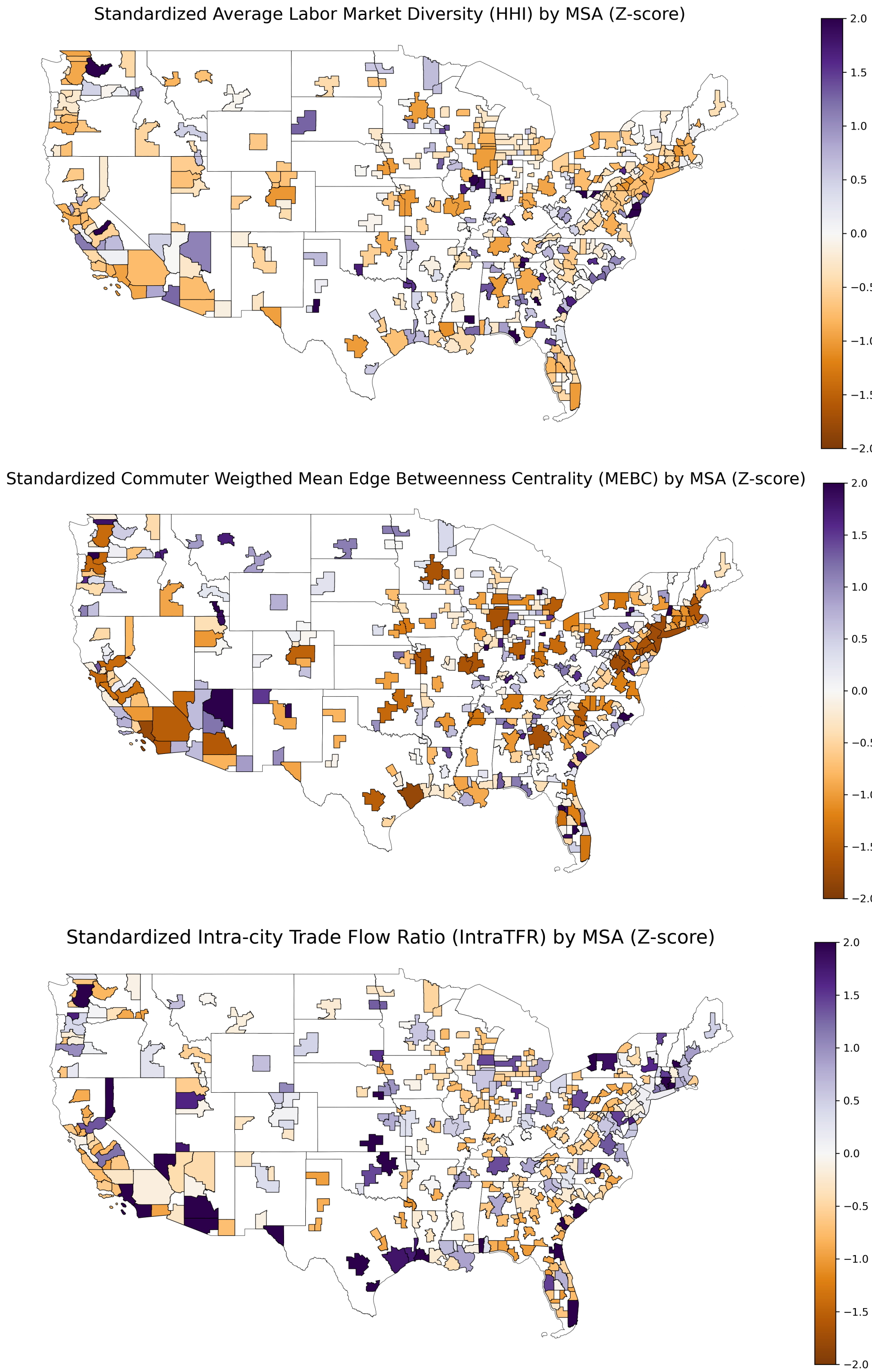



The strongest pairwise relationship is a moderate negative correlation of $-0.46$ between $wMEBC$ and $IntraTFR$. Despite both having a transportation-related dimension, the two measures capture different margins of urban structure. $wMEBC$ reflects the physical criticality of road connections, with greater weight placed on central road segments associated with workplaces requiring physical presence, whereas $IntraTFR$ captures the geographic scope of goods sourcing, which can also reflect the industries and suppliers available within the local economy. An MSA can therefore have a relatively distributed road network while still exhibiting substantial within-city goods sourcing.

The physical concentration of critical road connections and the geographic scope of goods sourcing need not move together.

Figure 7 presents the MSA-level standardized pre-period measures of city economic structures, enabling direct comparisons across cities. The raw measures are shown in Appendix Figure 14. The standardized geographic distributions of these measures show substantial heterogeneity across MSAs.

Figure 7: City-Level Concentration Measures (2017): Standardized Z-Scores




*Note*: This figure presents the standardized Z-scores of average concentration measures for the labor market, road network, and trade perspectives across MSAs from 2017, allowing for cross-variable comparison.

# IV Research Design

## IV.I LP-DiD for Transient Natural Disasters

I begin by estimating how the economic effects of natural disasters vary with urban economic structure using monthly MSA-level variation in disaster exposure. I apply the Local Projection Difference-in-Differences (LP-DiD) method developed by Dube et al. (2023) and Jordà et al. (2025), focusing on its non-absorbing treatment setting. This framework provides a novel approach to accommodating the transient nature of natural-disaster exposure experienced by cities.

Unlike systematic changes such as long-lasting policy interventions, most natural disasters occur as isolated, short-lived events. Because of this temporary nature, MSAs may experience disasters repeatedly over time, moving into and out of disaster exposure from one month to another. Treatment status can therefore change repeatedly over the sample period rather than remaining fixed after an initial disaster. This on-and-off pattern of exposure makes it difficult to define persistent treated and untreated groups. For example, a disaster occurring in one month may be followed by another event two months later, causing a previously treated MSA to be exposed again after only a one-month interval. Likewise, an MSA that initially serves as a control may subsequently experience a disaster, preventing it from remaining untreated throughout the analysis period. In summary, treatment and control status evolves continuously over time. Consequently, conventional difference-in-differences specifications may generate problematic comparisons when repeated disaster episodes create overlapping treatment histories across cities and over time. Furthermore, staggered DiD estimators are also not well-suited to this setting because they typically assume that treatment, once received, persists over time, as is common for policy interventions (De Chaisemartin et al. (2020); Callaway et al. (2021); Goodman-Bacon (2021); Sun et al. (2021); Borusyak et al. (2024)).

The non-absorbing LP-DiD framework addresses this issue by constructing treatment and control observations around each new treatment episode. An MSA is considered validly treated in month $t$ if it experiences a natural disaster in that month but was not exposed in the preceding month, $t-1$. If the same MSA subsequently becomes unexposed, it is no longer considered treated simply because of its past disaster exposure. Instead, if the MSA experiences another disaster in a later month after being unexposed in the immediately preceding month, it can enter treatment again as a newly exposed MSA.

Control observations are required to remain unexposed to a natural disaster over the clean comparison window beginning at $t-2$. Relative to the treatment definition above, this starting point allows an MSA contributing a control observation to have the most flexible prior exposure history consistent with a clean comparison. In particular, an MSA contributing a control observation may have experienced a disaster as recently as $t-3$, but must remain unexposed from $t-2$ through the corresponding post-treatment horizon $t+h$. This restriction assumes that one month without further disaster exposure is sufficient for the effects of a previous event to stabilize before the comparison period begins at $t-2$. As a robustness check, I extend the stabilization period from one month to six months and find highly consistent results across alternative stabilization lengths, as reported in Appendix XI.VII.3. Namely, I retain treated MSA-month observations only when the MSA remains exposed to natural disasters for the entirety of the relevant post-treatment window, and clean-control MSA-month observations only when the MSA remains unexposed over the corresponding comparison window. This construction excludes intermittent changes in disaster exposure within each relevant window without permanently assigning an MSA to either the treatment or control group over the full sample period.

## IV.II Econometric Framework

Building on this LP-DiD design, I estimate how the effect of disaster exposure varies across the three dimensions of urban economic structure by interacting each measure with treatment status. The resulting specification is as follows:

$$
\begin{aligned}
\ln(Load)_{c,m+h,y} - \ln(Load)_{c,m-1,y} = {} & (\delta_{m+h,y} - \delta_{m-1,y}) \\
& + \alpha_1\, \Delta D_{c,m,y} \\
& + \Delta D_{c,m,y}\Big[\underbrace{\beta_1\, HHI_c}_{\substack{\text{Labor Mkt:}\\ \text{ind. comp.}}} + \underbrace{\beta_2\, wMEBC_c}_{\substack{\text{Road Net.:}\\ \text{central.}}} + \underbrace{\beta_3\, IntraTFR_c}_{\substack{\text{Trade Flow:}\\ \text{goods sourcing}}} + \underbrace{\beta_4\, PopDensity_c}_{\substack{\text{Overall Intensity:}\\ \text{Pop density}}}\Big] \\
& + \gamma_1 HDD_{c,m+h,y} + \gamma_2 HDD_{c,m-1,y} + \gamma_3 CDD_{c,m+h,y} + \gamma_4 CDD_{c,m-1,y} \\
& + \gamma_5 OutageCount_{c,m+h,y} + \gamma_6 OutageCount_{c,m-1,y} + \gamma_7 BuildingSh_{\text{Pre-1980},c,y} \\
& + \epsilon_{c,m+h,y} \qquad (9)
\end{aligned}
$$

In Equation 9, $c$ denotes the MSA, $m$ denotes the treatment month, $h \geq 0$ denotes the post-disaster horizon in months, and $y$ denotes the calendar year from 2018 to 2024. The outcome variable, $\ln(Load)_{c,m+h,y} - \ln(Load)_{c,m-1,y}$, captures the change in logged industrial and commercial electricity consumption $h$ periods after a disaster relative to its level in the month preceding the disaster.

This outcome specification is particularly useful for electricity consumption, which exhibits substantial differences in baseline consumption levels across cities. Expressing the outcome relative to each MSA's own pre-disaster level removes persistent differences in consumption levels across MSAs. Furthermore, electricity consumption generally exhibits strong seasonality across months. To account for this seasonality, I additionally control for heating and cooling degree days, $HDD$ and $CDD$, in both the pre-disaster and outcome periods. Consistent with this long-difference outcome, $\epsilon_{c,m+h,y}$ denotes the error term in the long-difference specification. Throughout the paper, I cluster standard errors at the MSA level to account for serial correlation in the error terms within MSAs over time.

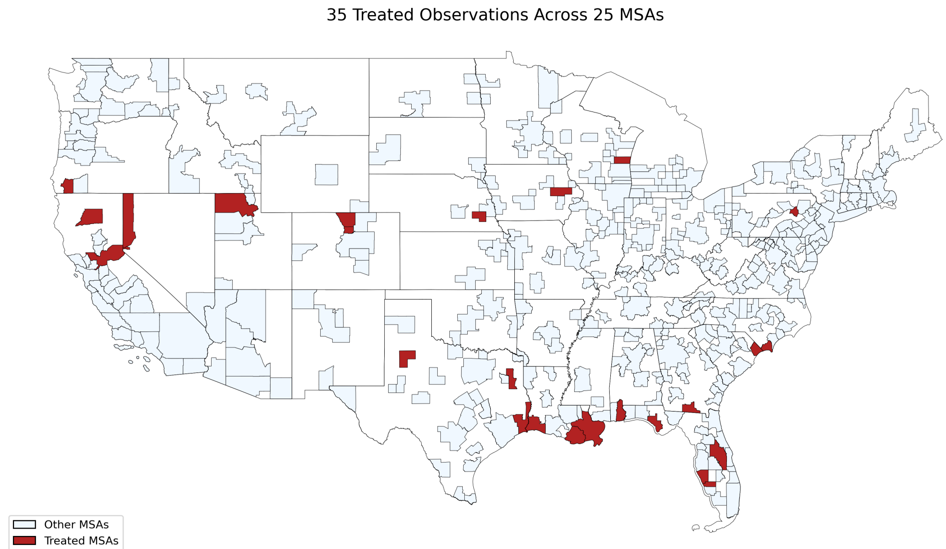
The treatment status variable $\Delta D_{c,m,y}$ equals one if MSA $c$ is newly exposed in month $m$ to a high-wind natural disaster, defined by an average wind intensity exceeding 96 miles per hour across the disaster events affecting the MSA in that month. Because treatment is non-absorbing, an MSA may enter treatment more than once over the sample period, subject to the treatment criteria described above. I interact treatment with $PopDensity$ to account for heterogeneity associated with overall urban concentration intensity and agglomeration forces. The term $(\delta_{m+h,y} - \delta_{m-1,y})$ denotes the differenced month-by-year fixed effects following the LP-DiD framework, which absorb aggregate time-varying factors common across MSAs between the pre-disaster period and the corresponding post-disaster horizon, including common seasonal fluctuations and nationwide shocks. In addition to $HDD$ and $CDD$, I control for the number of power outages, $OutageCount$, in both the pre-disaster and outcome periods, as well as the share of buildings constructed before 1980, $BuildingSh_{\text{Pre-1980}}$. These two additional controls are intended to capture the direct physical effects of natural disasters on electricity supply and infrastructure, rather than changes in the underlying economic activity that electricity consumption is used to proxy. Appendix XI.I examines the role of these controls by comparing the contemporaneous treatment effect with and without the weather, outage, and building-stock controls included in the preferred specification.

## IV.III Disaster Exposure Across MSAs

The analysis covers 365 MSAs over the 2018–2024 period. Applying the treatment criteria described above identifies 35 treated MSA-month observations across 25 unique MSAs. Figure 8 presents the geographic distribution of these treated observations. As expected, treated MSAs are concentrated along coastal regions, particularly the Gulf Coast, although the sample also includes several inland

MSAs.

Figure 8: Geographic Distribution of Hit MSAs (2018-2024)




*Note*: The figure shows MSAs affected by natural disasters between 2018 and 2024 that experienced Category 2+ high-wind events (Saffir–Simpson scale). These MSAs are the treated units in the LP-DiD design.

Table 3 compares the characteristics of treated and clean-control MSA-month observations entering the LP-DiD estimation sample in the corresponding treatment month. The table reports electricity consumption, the three measures of urban economic structure, population density, and climatological conditions in the reference month.

A remaining consideration is whether treated MSA-month observations are drawn from MSAs with systematically different prior exposure to natural disasters than those contributing clean-control observations. To assess this possibility, Table 4 compares wind exposure over the 2007–2016 pre-period between treated and clean-control MSA-month observations entering the estimation sample. Treated observations exhibit somewhat higher pre-period maximum wind speeds, whereas clean-control observations exhibit slightly higher pre-period average wind speeds. However, neither difference is statistically significant, providing little evidence of systematic differences in prior wind exposure between the two sets of observations.

Table 3: Summary Statistics for Treated and Clean-Control MSA-Month Observations

| | Treated Cities | | Control Cities | | | |
|---|---|---|---|---|---|---|
| Variable | Mean | SD | Mean | SD | Diff. | p-value |
| Log Electricity Sales | 12.622 | 1.222 | 12.118 | 1.274 | -0.504 | 0.019 |
| Population Density | 96.804 | 65.838 | 107.091 | 121.414 | 10.287 | 0.616 |
| Labor Market Concentration | 0.041 | 0.018 | 0.044 | 0.025 | 0.003 | 0.463 |
| Road Network Centrality (Commuting Weighted) | 0.013 | 0.007 | 0.014 | 0.006 | 0.001 | 0.491 |
| Within-City Goods Sourcing Ratio | 0.258 | 0.213 | 0.191 | 0.172 | -0.067 | 0.022 |
| $CDD$ (thousands) | 0.558 | 0.886 | 0.371 | 0.848 | -0.187 | 0.193 |
| $HDD$ (thousands) | 0.703 | 1.061 | 1.040 | 2.010 | 0.337 | 0.321 |
| Obs (MSA-months) | 35 | | 30,551 | | | |

*Note:* Summary statistics are reported at the MSA-month level for the $h = 0$ estimation sample, separately for treated observations entering high-wind disaster exposure and eligible clean-control observations. The table reports electricity sales, urban economic structure measures, population density, and climate controls in the reference month. Because treatment is non-absorbing, MSAs are not permanently assigned to the treatment or control group. The difference column reports Control minus Treated.

Table 4: Pre-period Wind Exposure across Treated and Control MSA-Month Observations

| | Treated | | Control | | | |
|---|---|---|---|---|---|---|
| Variable | Mean | SD | Mean | SD | Diff. | p-value |
| Pre-period Maximum Wind Speed | 58.1143 | 29.4267 | 53.1535 | 23.3198 | -4.9607 | 0.2086 |
| Pre-period Average Wind Speed | 16.8147 | 13.3796 | 19.6913 | 15.5623 | 2.8766 | 0.2744 |
| Obs (MSA-months) | 35 | | 30,467 | | | |

*Note:* This table reports pre-period hurricane exposure measures for the estimation sample at the MSA-month level. “Pre-period Maximum Wind Speed” is the maximum observed wind speed in a given MSA-month during the pre-period, while “Pre-period Average Wind Speed” is the corresponding average wind speed. The treated sample consists of 35 MSA-month observations contributed by 25 unique MSAs. The difference column reports Control minus Treated. P-values are from two-sample t-tests with equal variances.

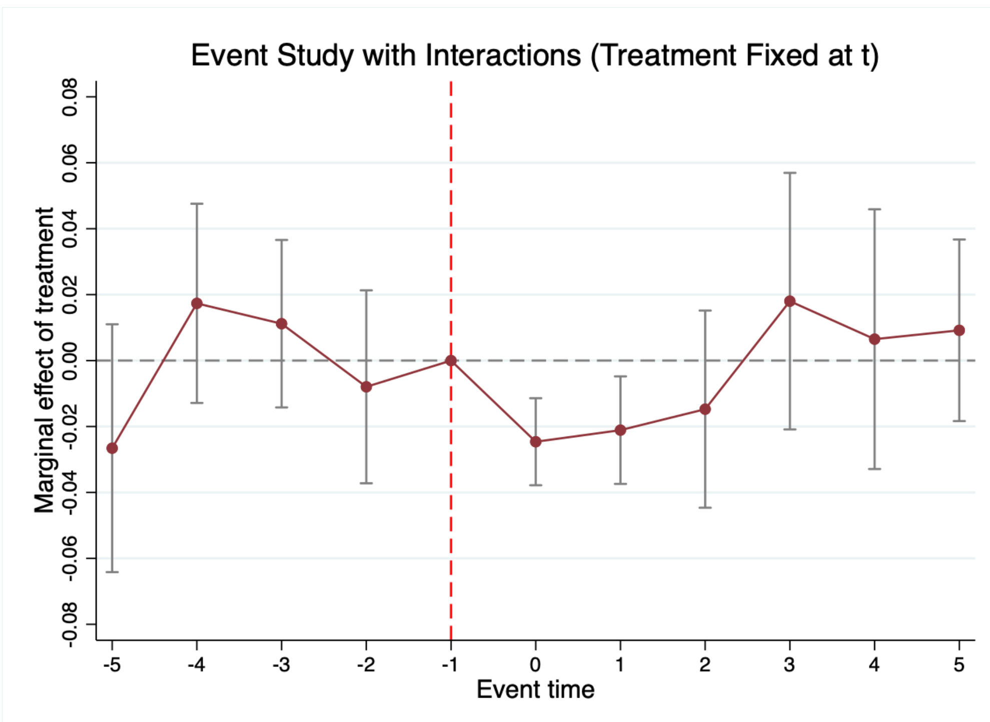
# V Results

## V.I Dynamic Responses to Natural Disaster Hits

I begin by examining the dynamic response of electricity consumption to natural disaster hits. An important question is whether changes in the economic disruption is concentrated in the event month or persists into subsequent periods. Figure 9 presents the corresponding event-study estimates. To isolate the dynamic response to a single disaster episode, I construct the event study around one-period disaster shocks. Treated observations experience a qualifying disaster hit in month $t$ but no additional qualifying disaster hits over the subsequent five months, while clean-control observations remain unexposed throughout the corresponding event window.[12] Accordingly, the estimated coefficients capture the dynamic response following an isolated natural disaster hit.

Figure 9 indicates that the economic disruption induced by natural disaster exposure is highly transitory. The largest decline in electricity consumption occurs during the event month ($h = 0$), with economically meaningful effects persisting into the immediately following month ($h = 1$). Beyond one month, however, the estimated effects attenuate considerably and become statistically indistinguishable from zero. This pattern motivates the focus of the subsequent analysis on the contemporaneous effect, which constitutes the primary estimand throughout the remainder of the paper.

Figure 9: Event-study with treatment assigned at the onset month only




*Note*: The figure reports event-study estimates of the effect of natural disaster hits on electricity consumption using isolated, one-period treatment shocks with clean-window controls. Event time is measured in months relative to the disaster hit ($h = 0$). The omitted category is the month immediately preceding the disaster hit ($h = -1$). Vertical bars denote 95% confidence intervals based on standard errors clustered at the city level.

[12] At the contemporaneous horizon ($h = 0$), the event-study sample coincides with the baseline non-absorbing LP-DiD specification. For post-treatment horizons ($h > 0$), the dynamic specification instead follows an isolated disaster occurring at $t$ and excludes treated observations experiencing another qualifying disaster during the subsequent event window. This construction allows the post-event coefficients to trace the evolution of electricity consumption following a single disaster episode rather than the effect of remaining continuously exposed over multiple periods.

## V.II Baseline Results

Having established the dynamic response to natural disaster hits, I now focus on their contemporaneous economic impact. Table 5 reports the baseline treatment effect, evaluated at the sample means of all urban-structure characteristics and population density. The estimated coefficient of $-0.0287$ implies that monthly electricity load declines by approximately 2.8 percent in the event month relative to the preceding month. Panel B reports the interactions between natural disaster exposure and the urban-structure characteristics. All three structural interactions are statistically significant, indicating substantial heterogeneity around the baseline effect. Greater industrial concentration significantly amplifies disaster-related economic losses, whereas stronger road connectivity and greater intra-city goods sourcing mitigate them. Population density also exhibits a negative and statistically significant interaction with disaster exposure, indicating larger losses in more densely populated cities. The economic magnitude of these differences is examined below.

Table 5: Contemporaneous Effects of Natural Disaster Exposure and Treatment-Effect Heterogeneity

| | | Change in Monthly MSA Electricity Load |
|---|---|---|
| | | Event Month ($h = 0$) |
| *Panel A: Average Treatment Effect* | | |
| | Average treatment effect | $-0.0287^{***}$ (0.0056) |
| *Panel B: Interaction Coefficients* | | |
| Labor Market | Industrial Composition ($HHI$) | $-1.585^{***}$ (0.288) |
| Road Network | Road Connectivity ($wMEBC$) | $3.049^{***}$ (0.939) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.0745^{***}$ (0.0275) |
| Demographics | Population Density | $-0.000422^{***}$ (0.0000872) |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and population density. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. The specification controls for heating degree days (HDD), cooling degree days (CDD), and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. Standard errors, clustered at the MSA level, are reported in parentheses. MSA-level electricity load is constructed using population-based weights. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

To quantify the economic magnitude of this heterogeneity, I evaluate the marginal treatment effect at one standard deviation above and below the sample mean of each structural measure, holding the others at their sample means, which together characterize the benchmark city. Table 6 reports the resulting treatment effects across these low- and high-intensity cases. For industrial composition, the heterogeneity is particularly pronounced. At one standard deviation above the mean of industrial concentration ($HHI$), a natural disaster is associated with an approximately 6.6 percent reduction in economic activity, compared with roughly 2.8 percent for the benchmark MSA. Thus, greater industrial concentration more than doubles the estimated economic loss from a disaster, with the treatment effect approximately 2.3 times as large as the benchmark effect. This represents the strongest gradient among the three dimensions of urban structure considered.

In contrast, greater road connectivity substantially mitigates the adverse economic impact of natural disasters. A one-standard-deviation increase in network centrality ($wMEBC$) reduces the magnitude of the estimated treatment effect by approximately 67 percent relative to the benchmark, indicating that cities with greater concentration of commuting-weighted connectivity along critical road links experience smaller contemporaneous economic losses following a disaster. At this higher level of road connectivity, the estimated treatment effect is also no longer statistically distinguishable from zero. A similar pattern emerges for intra-city goods sourcing. A one-standard-deviation increase in sourcing concentration ($IntraTFR$) reduces the magnitude of the treatment effect by approximately 44 percent relative to the benchmark, suggesting that cities with more locally concentrated sourcing relationships are less adversely affected by disaster shocks. Together, these estimates reveal economically substantial heterogeneity.

## V.III Alternative Measures of Economic Activity

As a broad monthly measure of metropolitan economic activity, the baseline analysis uses commercial and industrial electricity consumption. Because electricity demand responds immediately to disruptions in business operations and reflects the intensity of commercial and industrial activity, it provides a timely and broad measure of economic disruptions following natural disasters. At the same time, disaster-induced economic adjustment may also occur along another dimension, such as the labor margin. To examine this additional dimension of adjustment, I consider the three-month average change in employment as a complementary measure of economic activity. This provides a complementary test of whether the patterns observed in electricity consumption persist across a distinct margin of local economic adjustment.

Table 6: Heterogeneous Treatment Effects by Urban Structure Intensity

| | | Treatment Effects Across Urban Structure Levels | | | Baseline Effect |
|---|---|---|---|---|---|
| | | Monthly Electricity Load | | | |
| Category | Variable | Low | High | Difference | |
| Labor Market | Industrial Composition ($HHI$) | 0.0109 (0.0097) | $-0.0684^{***}$ (0.0085) | $-0.0793^{***}$ (0.0144) | $-0.0287^{***}$ (0.0056) |
| Road Network | Road Connectivity ($wMEBC$) | $-0.0481^{***}$ (0.0096) | $-0.0094$ (0.0063) | $0.0387^{***}$ (0.0119) | |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $-0.0415^{***}$ (0.0095) | $-0.0159^{***}$ (0.0039) | $0.0256^{***}$ (0.0094) | |
| Observations: 30,221 | | | | | |

*Notes:* The Low and High columns report treatment effects evaluated at one standard deviation below and above the sample mean of each urban-structure measure, respectively, while holding the remaining urban-structure characteristics and population density at their sample means. The Difference column reports the difference between the High and Low treatment effects. The Baseline Effect column reports the treatment effect evaluated with all urban-structure characteristics and population density fixed at their sample means. The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Standard errors are clustered at the MSA level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

### V.III.1 Three-Month Employment Changes

I focus on changes in employment rather than employment levels because the two measures capture different aspects of short-run labor market adjustment. Short-run disruptions may be difficult to detect in employment levels, which reflect the existing stock of workers. For instance, following a temporary disruption, employers may initially slow or postpone planned hiring without immediately reducing their existing workforce. Changes in employment are therefore better suited to capturing short-run disruptions to labor market adjustment following natural disasters.

I use the three-month average change in employment as the labor market outcome from the Current Employment Statistics–State and Area (CES–SAE) dataset[13]. The three-month window follows the CES practice of reporting three-month average employment changes, which helps reduce sampling variability in month-to-month estimates from the establishment survey. At the same time, the relatively short window preserves the measure's ability to capture near-term labor market adjustment following a disaster. Table 7 reports the event-month results using the CES–SAE three-month average employment change. Natural disaster exposure reduces the three-month average monthly employment change by approximately 576 workers relative to the corresponding counterfactual change. In other words, employment growth in the event month is lower by approximately

[13]Specifically, the three-month average change in employment for metropolitan area $c$ in month $t$ is defined as $\Delta Emp^{(3)}_{c,t} = \frac{1}{3}\sum_{k=0}^{2}\left(Emp_{i,t-k} - Emp_{c,t-k-1}\right)$, where $Emp_{ct}$ denotes CES employment (in thousands of employees). Positive values indicate employment expansion, while negative values indicate employment contraction.

576 workers per month than it would have been in the absence of disaster exposure. For scale, treated MSAs experienced an average monthly employment increase of approximately 1,064 workers prior to disaster exposure, so the estimated reduction is equivalent to roughly 54 percent of their average pre-disaster employment growth. This result complements the decline in commercial and industrial electricity consumption documented in the baseline analysis, showing that disaster-induced disruptions are also reflected in firms' short-run employment decisions, consistent with delayed or reduced hiring following a disruption. The heterogeneity in employment responses further supports the main findings based on commercial and industrial electricity consumption. As reported in Appendix Table 18, the decline in employment growth is larger in MSAs with greater labor-market concentration, but smaller in MSAs with greater road connectivity and stronger intra-city trade networks, consistent with the corresponding heterogeneity in the baseline results. Thus, the role of urban structure extends beyond electricity consumption to short-run labor market adjustment.

Table 7: Contemporaneous Effects of Natural Disaster Exposure on the Three-Month Average Employment Change

| | | Three-Month Average Employment Change |
|---|---|---|
| | | Event Month ($h = 0$) |
| *Panel A: Average Treatment Effect* | | |
| | Average treatment effect | $-0.574^{***}$ (0.220) |
| *Panel B: Interaction Coefficients* | | |
| Labor Market | Industrial Composition ($HHI$) | $-14.903$ (11.286) |
| Road Network | Road Connectivity ($wMEBC$) | $48.225^{**}$ (19.525) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $1.551^{***}$ (0.564) |
| Demographics | Population Density | $-0.00987^{***}$ (0.00310) |
| | Observations | 30,436 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Standard errors, clustered at the MSA level, are reported in parentheses. Electricity-specific controls, including heating degree days, cooling degree days, outage measures, and the share of pre-1980 housing stock, are excluded. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

# VI Industry-Specific Heterogeneity in Disaster Vulnerability

Results from Section V.II show that various urban economic structures can either amplify or mitigate the economic impacts of natural disasters. Among the three dimensions considered, industrial concentration exhibits the strongest heterogeneity, with greater concentration substantially amplifying disaster-induced economic losses. This aggregate relationship, however, does not reveal which parts of the local industrial structure generate this vulnerability. Because the $HHI$ summarizes concentration across all industrial subsectors, the estimated effect may reflect broad-based industrial concentration or instead be driven disproportionately by particular industry groups.

To distinguish between these possibilities, SectionVI decomposes overall industrial concentration into six industry group–specific $HHI$ measures and examines their respective contributions to disaster vulnerability. The six industry groups comprise Retail, Utilities, and Construction (RUC), Health and Education (HE), Professional and Business Services (PS), Manufacturing (M), Accommodation, Transportation, and Wholesale Trade (ATT), and Other Industries. This extension also speaks to the well-established literature attributing recent urban growth primarily to the expansion and specialization of professional and business service industries, which have driven much of the recent productivity gains[14] (Caliendo et al. (2018); Eckert et al. (2022); Rossi-Hansberg et al. (2021)). The analysis, therefore, distinguishes whether the aggregate relationship reflects broad-based industrial concentration across the local economy or is driven disproportionately by concentration within specific industry groups.

## VI.I Industry Group–Specific $HHI$ Measures

To identify which parts of the local industrial structure contribute to the greater disaster-induced losses under higher overall industrial concentration, I decompose the baseline $HHI$ into six industry group–specific components. For industry group $g$ in MSA $c$, I define

$$HHI_{gc} = \sum_{i \in g} \left( \frac{Emp_{ic}}{TotalEmp_c} \right)^2, \qquad g \in \{\text{RUC}, \text{HE}, \text{PS}, \text{M}, \text{ATT}, \text{Other}\} \tag{10}$$

[14] The professional and business services sector includes NAICS codes 51–56, comprising Information (51), Finance and Insurance (52), Real Estate and Rental and Leasing (53), Professional, Scientific, and Technical Services (54), Management of Companies and Enterprises (55), and Administrative and Support and Waste Management and Remediation Services (56). For instance, Eckert et al. (2022) show that between 1980 and 2015, urban-biased wage growth originated almost entirely within professional and business services, where the wage gap between high- and low-density areas widened substantially while remaining largely flat in other sectors.

where $Emp_{ic}$ denotes employment in industry $i$ and $TotalEmp_c$ denotes total employment in MSA $c$. The summation is restricted to industries belonging to group $g$, while employment shares continue to be defined relative to total MSA employment. Thus, $HHI_{gc}$ captures the contribution of industries within group $g$ to the MSA's overall industrial concentration, rather than concentration within the group conditional on its own employment.

Because the six industry groups partition the industries entering the baseline concentration measure, the overall $HHI$ can be written as the sum of these group-specific components:

$$\begin{aligned} HHI_c &= \sum_g HHI_{gc} \\ &= HHI_{RUC,c} + HHI_{HE,c} + HHI_{PS,c} + HHI_{M,c} + HHI_{ATT,c} + HHI_{Other,c} \end{aligned} \tag{11}$$

## VI.II Industry Group Concentration Profiles of Treated MSAs

I begin by characterizing the industry group concentration profiles of treated MSAs across the six groups defined above. These profiles show how different industry groups contribute to overall industrial concentration across disaster-exposed cities. The measures are constructed from each MSA's average industry composition over the pre-treatment period from 2013 to 2017, thereby capturing persistent differences in local industrial structure rather than contemporaneous responses to disaster exposure.

Table 8 reports the distribution of treated MSAs across high, average, and low concentration categories for each industry group. The industry-group-specific HHIs are standardized across all MSAs and classified into quartiles, with the top quartile labeled "High", the middle two quartiles "Average", and the bottom quartile "Low". It reveals substantial heterogeneity in industry-group concentration across cities. First, the industry groups in which MSAs exhibit high concentration differ considerably across cities, indicating that similar levels of overall industrial concentration can reflect very different patterns of industry specialization. Second, several MSAs exhibit high concentration in multiple industry groups rather than in a single group. For example, Ogden–Clearfield, UT is highly concentrated in RUC, HE, PS, and M, while Beaumont–Port Arthur, TX exhibits high concentration in RUC, HE, M, and ATT. These patterns show that local industrial specialization can span multiple industry group and the importance of distinguishing among its industry-group-specific components when assessing heterogeneity in disaster impacts. Figure 10 provides a more concise visualization of these profiles by displaying, for each of the 25 treated

MSAs, only the industry groups classified as highly concentrated.

Table 8: Industry-group concentration profiles of treated MSAs

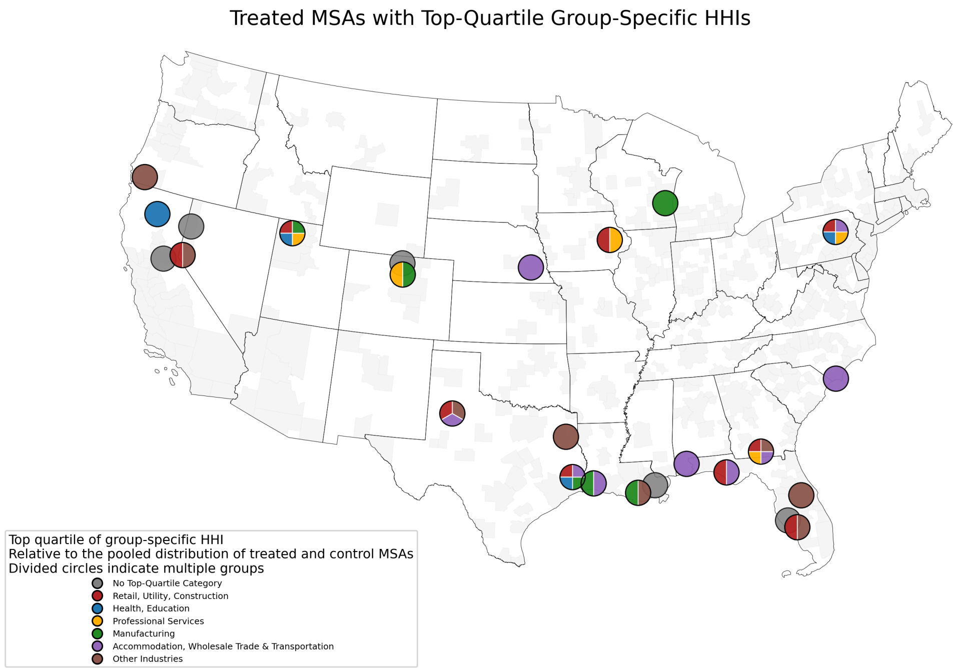
| MSA | RUC | HE | PS | Manu | ATT | Others |
|---|---|---|---|---|---|---|
| Beaumont-Port Arthur, TX | High | High | Low | High | High | Low |
| Bloomsburg-Berwick, PA | High | High | High | Avg | High | Avg |
| Boulder, CO | Low | Avg | High | High | Avg | Avg |
| Carson City, NV | High | Avg | Avg | Avg | Avg | High |
| Cedar Rapids, IA | High | Avg | High | Avg | Avg | Low |
| Daphne-Fairhope-Foley, AL | Avg | Low | Low | Avg | High | Avg |
| Fort Collins, CO | Avg | Low | Avg | Avg | Avg | Avg |
| Grants Pass, OR | Avg | Avg | Avg | Avg | Avg | High |
| Houma-Thibodaux, LA | Avg | Low | Avg | High | Avg | High |
| Lake Charles, LA | Avg | Avg | Avg | High | High | Avg |
| Lincoln, NE | Avg | Low | Avg | Avg | High | Avg |
| Longview, TX | Avg | Avg | Avg | Avg | Avg | High |
| Lubbock, TX | High | Low | Avg | Avg | High | High |
| Myrtle Beach-Conway-North Myrtle Beach, SC-NC | Avg | Low | Avg | Low | High | Low |
| New Orleans-Metairie, LA | Low | Avg | Avg | Avg | Avg | Avg |
| North Port-Sarasota-Bradenton, FL | Avg | Avg | Avg | Avg | Avg | Avg |
| Ogden-Clearfield, UT | High | High | High | High | Low | Avg |
| Orlando-Kissimmee-Sanford, FL | Avg | Avg | Avg | Low | Avg | High |
| Panama City, FL | High | Avg | Avg | Avg | High | Low |
| Punta Gorda, FL | High | Avg | Low | Low | Avg | High |
| Redding, CA | Avg | High | Avg | Avg | Avg | Avg |
| Reno, NV | Avg | Avg | Avg | Avg | Avg | Avg |
| Sacramento–Roseville–Arden-Arcade, CA | Avg | Avg | Avg | Low | Avg | Avg |
| Sheboygan, WI | Avg | Avg | Avg | High | Avg | Avg |
| Valdosta, GA | High | Avg | High | Avg | High | High |

Note: High, Avg, and Low denote the upper, middle, and lower quartiles of the standardized industry-group concentration measures, respectively. The measures are standardized across all MSAs using average industry composition over the 2013–2017 pre-treatment period.

## VI.III Results Using Industry Group–Specific $HHI$ Measures

Building on the substantial variation in industry-group concentration documented across treated MSAs, I extend the baseline specification as shown in Equation 12, relaxing the restriction that concentration across different industry groups has a common effect on disaster impacts. Whereas the baseline specification summarizes industrial concentration using the aggregate measure $HHI_c$, the extended specification enters each industry group–specific component, $HHI_{gc}$, separately and allows it to have its own interaction coefficient. This decomposition therefore reveals whether the

Figure 10: Geographic Distribution of Industry-Group Specialization Across Treated MSAs




amplification of disaster-induced losses under greater overall industrial concentration is broadly shared across industry groups or driven by concentration in particular parts of the local economy.

$$
\begin{aligned}
\ln(Load)_{c,m+h,y} - \ln(Load)_{c,m-1,y} = {} & (\delta_{m+h,y} - \delta_{m-1,y}) + \alpha_1 \Delta D_{c,m,y} \\
& + \Delta D_{c,m,y} \times \Big[\beta_{11} HHI_{RUC,c} + \beta_{12} HHI_{HE,c} \\
& \qquad + \beta_{13} HHI_{BS,c} + \beta_{14} HHI_{MANU,c} \\
& \qquad + \beta_{15} HHI_{ATT,c} + \beta_{16} HHI_{OTHER,c}\Big] \\
& + \Delta D_{c,m,y} \times \Big[\beta_2 wMEBC_c + \beta_3 IntraTFR_c + \beta_4 PopDensity_c\Big] \\
& + \gamma_1 HDD_{c,m+h,y} + \gamma_2 HDD_{c,m-1,y} + \gamma_3 CDD_{c,m+h,y} + \gamma_4 CDD_{c,m-1,y} \\
& + \gamma_5 OutageCount_{c,m+h,y} + \gamma_6 OutageCount_{c,m-1,y} + \gamma_7 BuildingSh_{\text{Pre-1980},c,y} \\
& + \epsilon_{c,m+h,y} \qquad (12)
\end{aligned}
$$

Table 9 reports ports the estimated heterogeneity in disaster impacts across the industry-group-specific concentration measures and reveal substantial heterogeneity. In particular, concentration in manufacturing appears to mitigate the economic consequences of natural disasters, with estimated effects being highly statistically significant. In contrast, concentration in the HE and ATT industry groups exacerbates the economic consequences of natural disasters. Overall, these results demonstrate that the aggregate exacerbating effect identified in the baseline specification masks substantial sectoral heterogeneity.

Table 9: Marginal Effects in the Event Month ($h = 0$): Industry Group–Specific $HHI$ Measures

| | | Change in Monthly MSA Electricity Load |
|---|---|---|
| | | Event Month ($h = 0$) |
| *Panel A: Treatment Effect at Sample Means* | | |
| | Treatment effect at sample means | $-0.0079^{*}$ (0.0041) |
| *Panel B: Interaction Coefficients* | | |
| Labor Market | Retail, Utilities, and Construction ($HHI_{\text{RUC}}$) | $-1.704$ (1.355) |
| | Health and Education ($HHI_{\text{HE}}$) | $-2.521^{**}$ (1.022) |
| | Professional Services ($HHI_{\text{PS}}$) | 0.578 (1.638) |
| | Manufacturing ($HHI_{\text{Manu}}$) | $3.553^{***}$ (1.027) |
| | Accommodation, Transportation, and Trade ($HHI_{\text{ATT}}$) | $-1.284^{***}$ (0.302) |
| | Other Industries ($HHI_{\text{Other}}$) | $5.135^{***}$ (1.586) |
| Road Network | Road Connectivity ($wMEBC$) | $2.035^{*}$ (1.060) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.0748^{***}$ (0.0258) |
| Demographics | Population Density | $-0.000371^{***}$ (0.0000614) |
| | Observations | 30,221 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all industry group–specific concentration measures, the remaining urban-structure characteristics, and population density. Panel B reports the estimated interaction coefficients between natural disaster exposure and each industry group–specific concentration measure, together with the remaining urban-structure characteristics. The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. Standard errors, clustered at the MSA level, are reported in parentheses. MSA-level electricity load is constructed using population-based weights. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Having established the heterogeneous roles of industry group–specific concentration , Table 10 illustrates how the estimated treatment effect varies across different levels of all industry group–specific $HHI$ measures ($HHI_{gc}$). The results reveal substantial differences across industry groups. Greater concentration in HE and ATT amplifies the economic consequences of natural disasters.

Moving from one standard deviation below to one standard deviation above the mean changes the estimated treatment effect from 0.0125 to −0.0282 for HE and from 0.0145 to −0.0302 for ATT, with both differences statistically significant. At the high-concentration level, the estimated disaster-induced reductions are approximately 3.57 and 3.82 times as large as the mean-evaluated effect of −0.0079 for HE and ATT, respectively. Taken together, these results suggest that the vulnerability-enhancing effect of greater overall industrial concentration is concentrated primarily in HE and ATT rather than broadly shared across industry groups.

An additional question raised by these results is what accounts for greater industry-group-specific concentration within an MSA. A higher industry group–specific $HHI_{gc}$ can reflect either a larger employment share of that industry group in the local economy or greater concentration of employment among a smaller number of industry subsectors within the group. Appendix XI.VI distinguishes between these two sources by constructing concentration measures based on employment shares within each industry group. This additional decomposition examines the role of within-group specialization separately from the relative scale of each industry group in the local economy.

Table 10: Heterogeneous Treatment Effects by Industry Group–Specific Concentration

| | Treatment Effects Across Industry Group Concentration | | | Effect at Mean |
|---|---|---|---|---|
| | Monthly Electricity Load | | | |
| Variable | Low | High | Difference | |
| Manufacturing ($HHI_{MANU}$) | $-0.0640^{***}$ (0.0173) | $0.0483^{***}$ (0.0162) | $0.1122^{***}$ (0.0325) | |
| Health & Education ($HHI_{HE}$) | 0.0125 (0.0087) | $-0.0282^{***}$ (0.0097) | $-0.0407^{**}$ (0.0165) | |
| Accommodation, Wholesale Trade & Transportation ($HHI_{ATT}$) | $0.0145^{***}$ (0.0050) | $-0.0302^{***}$ (0.0081) | $-0.0447^{***}$ (0.0105) | $-0.0079^{*}$ (0.0041) |
| Professional Services ($HHI_{PS}$) | −0.0109 (0.0099) | −0.0048 (0.0090) | 0.0060 (0.0171) | |
| Retail, Utilities, and Construction ($HHI_{RUC}$) | −0.0034 (0.0055) | $-0.0123^{**}$ (0.0054) | −0.0089 (0.0070) | |
| Other Industries ($HHI_{OTHER}$) | $-0.0703^{***}$ (0.0175) | $0.0546^{**}$ (0.0217) | $0.1248^{***}$ (0.0386) | |
| Observations: 30,221 | | | | |

*Notes:* The Low and High columns report treatment effects evaluated at one standard deviation below and above the sample mean of the corresponding industry group–specific concentration measure, respectively, while holding the remaining industry group–specific concentration measures, other urban-structure characteristics, and population density at their sample means. The Difference column reports the change in the estimated treatment effect from the Low to the High concentration level. The Baseline Effect column reports the treatment effect evaluated with all industry group–specific concentration measures, remaining urban-structure characteristics, and population density fixed at their sample means. The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Standard errors, clustered at the MSA level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

# VII Industry Mobility and Disaster Vulnerability

The industry-specific results in Section VI reveal substantial heterogeneity in how concentration across industry groups shapes the economic consequences of natural disasters. These findings characterize how individual components of an MSA's industrial composition contribute to its vulnerability to disaster shocks. Their implications may extend beyond the industries in which these vulnerabilities originate, however, once cross-industry connections are taken into account. If establishments in a particular industry are routinely connected through movements originating from any other industries, disruptions associated with that industry's vulnerability may have consequences beyond the economic activity occurring directly within the industry itself. To explore this broader dimension, I examine industry-to-industry mobility patterns based on individuals' routine movements between establishments. The purpose of this exercise is to characterize how the six industry groups considered in the preceding analysis are embedded in everyday urban activity and, in turn, to provide additional context for interpreting the heterogeneous industry-specific resilience results.

## VII.I Measuring Industry-to-Industry Mobility

I use device-level foot-traffic data from Veraset (2022) to construct industry-to-industry mobility networks. The data cover January 1, 2021 through September 29, 2022 and record sequences of establishment visits by individual mobile devices within an MSA over the course of a day. I order establishment visits within each device-day and identify transitions between consecutive stops, assigning each establishment to one of the six industry groups used in the previous section. Because the analysis focuses on mobility between economic establishments, I exclude transitions involving home. For example, for a device observed moving from home to an RUC establishment, then to a PS establishment, and subsequently returning home, the industry-to-industry network records the RUC-to-PS transition. To prevent device-days with longer observed itineraries from mechanically receiving greater weight, I normalize transitions within each device-day so that the weights assigned to all observed industry-to-industry transitions sum to one. I then aggregate these normalized transition weights within each MSA-month.

For each MSA-month, I construct the conditional transition probability

$$P_{ij,ct} = \frac{W_{ij,ct}}{\sum\limits_{k \in \mathcal{I}} W_{ik,ct}}, \qquad i, j \in \mathcal{I} \tag{13}$$

where $W_{ij,mt}$ denotes the aggregate normalized transition weight from origin industry $i$ to destination industry $j$ in MSA $c$ and month $t$, and $\mathcal{I}$ denotes the six industry groups. Thus, $P_{ij,ct}$ measures the probability that the next observed economic establishment belongs to industry group $j$, conditional on the current establishment belonging to industry group $i$. For example, $P_{\text{HE,ATT},ct} = 0.3$ indicates that, among normalized transitions from HE establishments to other economic establishments, 30 percent are directed toward ATT establishments. Finally, I average the monthly transition probabilities within each MSA and then across MSAs to obtain the aggregate industry-to-industry mobility network. The resulting $6 \times 6$ transition matrix therefore characterizes the industry-to-industry mobility pattern of the average MSA in the sample.

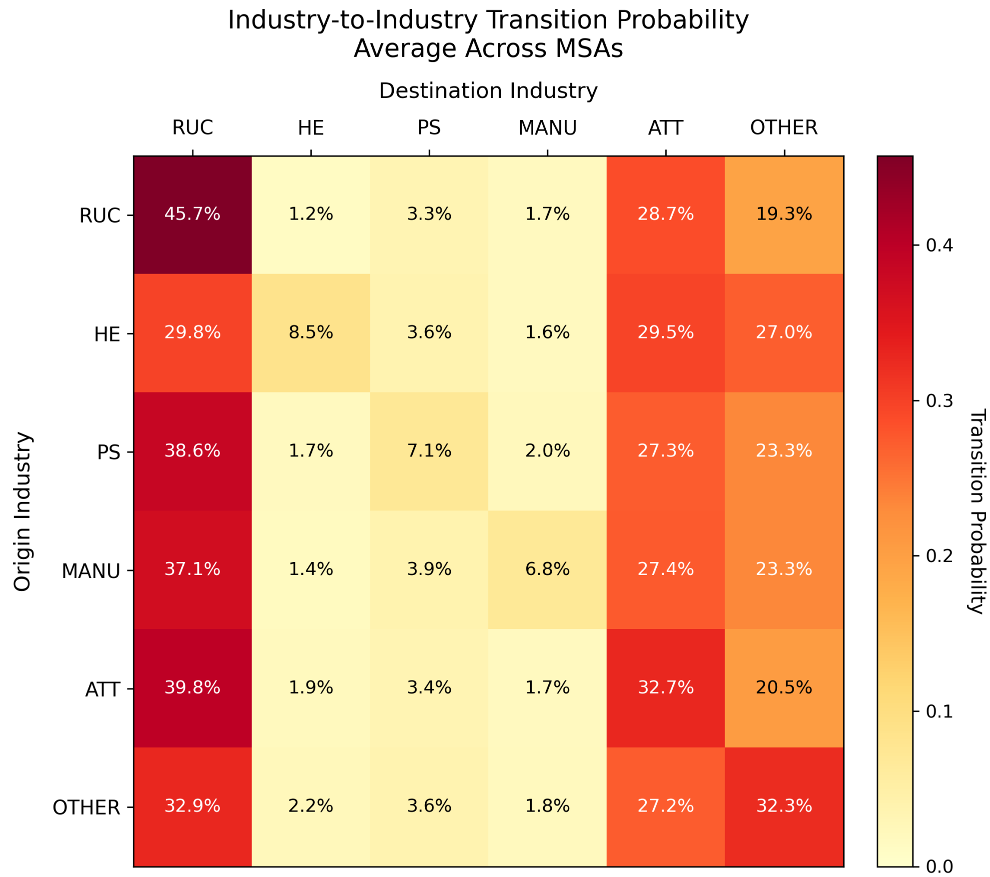
## VII.II Cross-Industry Mobility and the Vulnerability of Accommodation, Wholesale Trade, and Transportation

The resulting mobility network reveals substantial heterogeneity in how individual industry groups are embedded in routine movements across the city. As shown in Figure 11, RUC and ATT stand out as the two most prominent destinations, accounting for a substantial share of subsequent establishment visits regardless of the industry group from which a trip originates. This places RUC and ATT at the center of routine cross-industry mobility within a city. The prominence of ATT is particularly relevant to the industry-specific heterogeneity documented in Section VI, where greater ATT concentration substantially amplifies the economic consequences of natural disasters.

While Figure 11 summarizes the average industry-to-industry mobility pattern across all MSAs, I next examine whether the prominence of RUC and ATT as mobility destinations persists across MSAs with different patterns of industry concentration. More generally, the aggregate mobility pattern may mask systematic differences in industry-to-industry transitions across MSAs with different industrial compositions. I therefore examine how these mobility patterns vary with local industrial composition. For each of the six industry groups, I identify MSAs in the top quartile of the corresponding industry-specific $HHI$ measure, $HHI_{gc}$, and separately average their MSA-level transition matrices. This procedure yields six comparable mobility networks for MSAs characterized by high concentration in RUC, HE, PS, M, ATT, and Others, respectively.

Figure 12 confirms that the prominence of RUC and ATT as mobility destinations extends beyond MSAs concentrated in these industry groups. In particular, substantial mobility toward RUC and ATT is observed among MSAs with high concentration in Manufacturing, HE, PS, RUC, and ATT. The prominence of these industries in routine mobility therefore appears to be a broad

Figure 11: Industry-to-Industry Transition Probabilities Across Cities

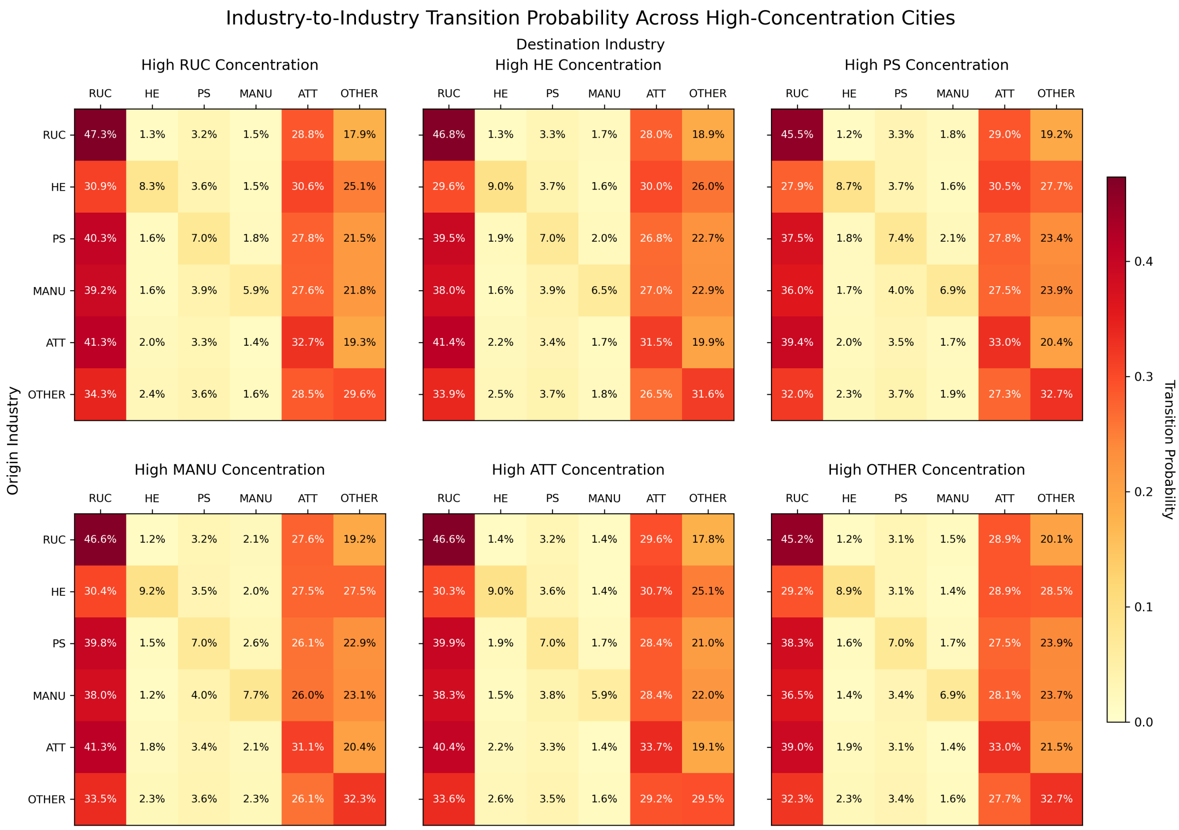



*Note*: This figure reports conditional transition probabilities across industry groups, excluding home-related transitions. Each cell shows the probability that the next observed non-home stop belongs to the destination industry indicated by the column, conditional on the current stop belonging to the origin industry indicated by the row. RUC and ATT emerge as the most prominent destinations across a wide range of origin industries.

feature of urban activity rather than one specific to cities concentrated in those industries.

These mobility patterns provide an additional perspective on the finding that greater ATT concentration exacerbates the economic consequences of natural disasters. ATT is a prominent destination for movements originating from nearly every industry group, indicating that ATT activities are closely connected to routine economic activity throughout the local economy. This feature is particularly informative when comparing ATT with HE. Although greater concentration in either ATT or HE amplifies disaster-induced losses, the two industry groups occupy notably different positions in the mobility network. ATT exhibits broad cross-industry mobility connections, whereas HE does not display the same pattern. The mobility evidence therefore points to cross-industry connectedness as a potentially important dimension of vulnerability for ATT-intensive economies, while suggesting that the greater vulnerability of HE-intensive economies may operate through other

Figure 12: Industry-to-Industry Transition Probabilities Across Highly Specialized Cities




*Note*: This figure reports industry-to-industry transition probabilities separately for cities in the top quartile of concentration in each industry group. City concentration is defined using the industry group–specific $HHI$ measure. The common transition structure is broadly preserved across concentration groups, with RUC and ATT remaining prominent destinations in each type of city.

dimensions of economic activity.

Viewed from this perspective, the mobility evidence also provides a way to think about the allocation of external resilience support. In ATT-intensive cities, public investments or post-disaster assistance that maintain or restore access to ATT-related activities may benefit economic activity beyond the directly affected industry group, given ATT's broad connections to routine mobility throughout the city.

# VIII Road Network Centralization and Disaster Resilience

## VIII.I Road Network Centralization and Traffic Delay During Weather Shocks

The baseline results show that greater road network centralization yields substantially smaller contemporaneous economic losses following natural disasters. To shed light on this mitigating relationship, I examine whether more centralized road networks are better able to maintain transportation activity during adverse weather conditions. The underlying idea is that centralized networks concentrate connectivity along a relatively small number of critical road segments, allowing the operation or restoration of these key corridors to preserve a larger share of overall network functionality during disruptions. This interpretation is consistent with transportation recovery practices. Federal Highway Administration (2013) emphasize that post-disaster traffic management prioritizes restoring "essential traffic" along key transportation corridors to facilitate emergency response and accelerate network recovery. Road network centralization may therefore contribute to disaster resilience by concentrating connectivity along critical links whose continued operation or rapid restoration helps limit transportation disruptions.

To assess whether more centralized road networks are associated with smaller traffic disruptions during adverse weather conditions, I relate road network centralization to weather-related vehicle hours of delay (VHD). Traffic performance measures are obtained from the Regional Integrated Transportation Information System (RITIS) Causes of Congestion database, which provides county-month measures of traffic delay relative to free-flow travel conditions for 2019. RITIS attributes congestion to several primary causes, including recurrent congestion, traffic incidents, work zones, holidays, weather, traffic signals, and combinations of these factors. VHD measures the cumulative hours of delay experienced by vehicles relative to free-flow travel, with one VHD corresponding to one vehicle experiencing one additional hour of delay. I focus specifically on weather-related VHD, which captures traffic delay attributed to adverse weather conditions such as rain, snow, fog, and ice. Lower weather-related VHD therefore indicates smaller traffic disruptions during adverse weather.

I compare the partial correlations between $wMEBC$ and different categories of VHD, controlling for metropolitan population size. Without this adjustment, more centralized road networks appear to be associated with lower traffic delay across all congestion categories, with little distinction between weather-related disruptions and other sources of congestion[15]. This broad pattern, however, may largely reflect differences in metropolitan scale and travel demand, as larger MSAs tend to exhibit both lower $wMEBC$ and higher levels of congestion. After controlling for metropolitan population size, Table 11 and Figure 13 reveal substantially different relationships across sources of traffic delay. In particular, the partial correlation between weather-related VHD and $wMEBC$ is negative, indicating that more centralized road networks are associated with lower weather-induced traffic delay after controlling for metropolitan population size. In contrast, the partial correlation

[15] Appendix Table 20 reports the unconditional correlations.

between recurrent-congestion VHD and $wMEBC$ is positive, consistent with routine traffic delay being greater when road connectivity is concentrated along a smaller number of major corridors.

These contrasting relationships highlight the potential resilience advantage of road network centralization during adverse weather, with more centralized networks exhibiting smaller weather-related traffic disruptions despite greater recurrent congestion under normal operating conditions. Viewed from this perspective, the road network results also provide a way to think about the spatial allocation of transportation resilience investments. Expanding road connectivity may not provide the same resilience benefits when additional connections are distributed across increasingly peripheral parts of the network. Instead, investments that maintain or strengthen connectivity along strategically important corridors may be particularly valuable for limiting transportation disruptions during natural disasters.

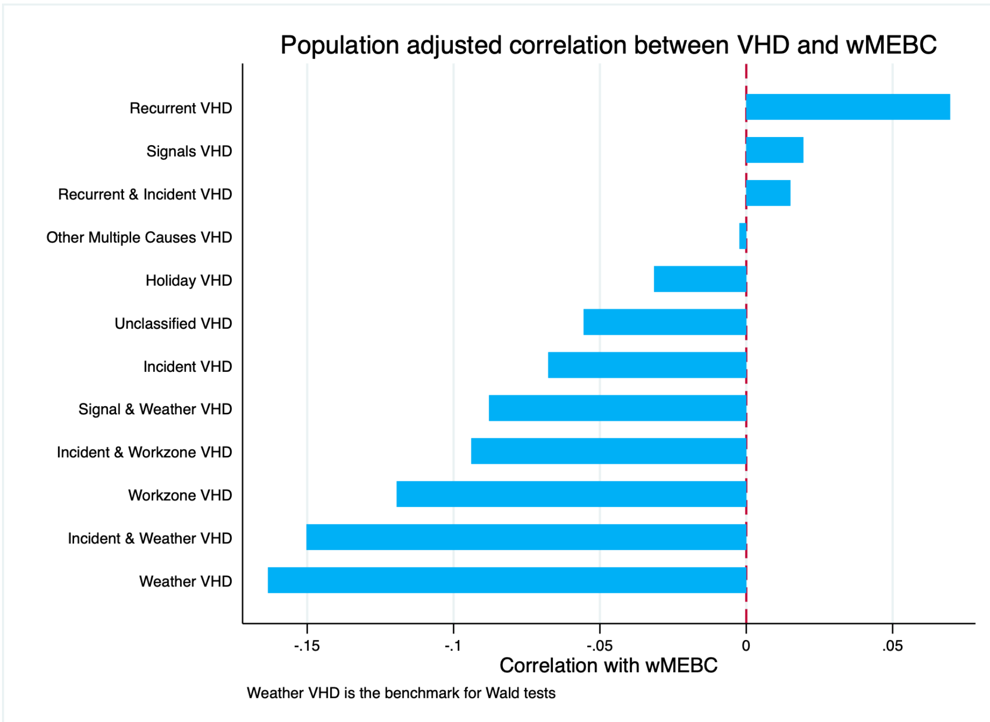
Table 11: Partial Correlations Between Weighted Mean Edge Betweenness Centrality and Vehicle Hours of Delay, Controlling for City Population

| VHD Category | Partial Correlation | Difference from Weather |
|---|---|---|
| Weather VHD | −0.163 | Reference |
| Incident & Weather VHD | −0.150 | |
| Workzone VHD | −0.119 | |
| Incident & Workzone VHD | −0.094 | * |
| Signal & Weather VHD | −0.088 | * |
| Incident VHD | −0.068 | ** |
| Unclassified VHD | −0.056 | ** |
| Holiday VHD | −0.032 | *** |
| Other Multiple Causes VHD | −0.002 | *** |
| Recurrent & Incident VHD | 0.015 | *** |
| Signals VHD | 0.020 | ** |
| Recurrent VHD | 0.070 | *** |

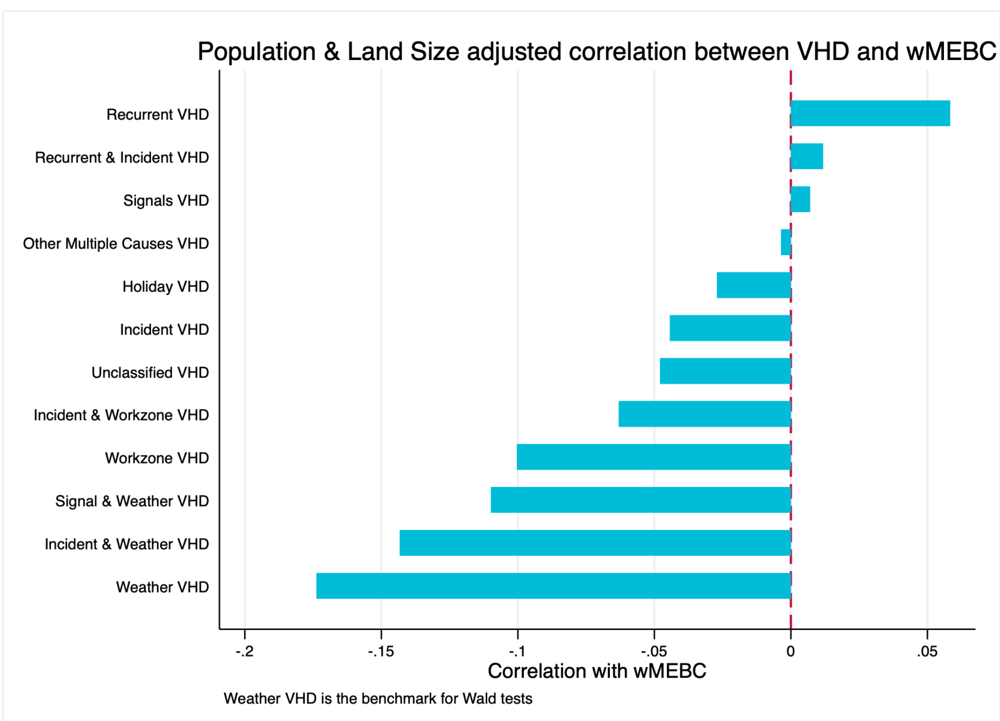
*Notes*: Entries report partial correlations between weighted Mean Effective Betweenness Centrality ($wMEBC$) and Vehicle Hours of Delay (VHD), estimated from a covariance structural equation model (SEM) controlling for city population. Weather VHD serves as the reference category. The "Difference from Weather" column reports the significance of Wald tests comparing each VHD–$wMEBC$ partial correlation with the Weather VHD–wMEBC partial correlation. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 13: Correlations between VHD categories and $wMEBC$ under alternative adjustment specifications

(a) Adjusted for population




(b) Adjusted for population and land area




*Note*: Bars represent Pearson correlation coefficients between each VHD category and $wMEBC$.

# IX Economic Costs of Disaster-Induced Electricity Reductions

This paper finds that natural disasters reduce a city's non-residential electricity consumption by approximately 2.8 percent on average. The analysis further shows that the magnitude of this reduction varies substantially across cities depending on their underlying urban structure. To provide a more tangible interpretation of these estimated effects, I translate the implied reductions in electricity consumption into monetary losses using interruption-cost estimates from the Interruption Cost Estimate (ICE) Calculator. The ICE Calculator, developed for the U.S. Department of Energy by Lawrence Berkeley National Laboratory and Freeman, Sullivan & Co., provides model-based estimates of the economic costs incurred by electricity customers during power interruptions. Because the empirical outcome in this paper measures non-residential electricity consumption, I use the corresponding non-residential interruption-cost estimates.

I value the estimated electricity reductions using the ICE measure of economic cost per unserved $kWh$, which directly expresses interruption costs relative to the amount of electricity not supplied[16]. This measure aligns directly with the empirical outcome in this paper, which captures reductions in non-residential electricity consumption. The 8-hour interruption scenario, corresponding to \$52 per unserved $kWh$, serves as the baseline valuation, while the 2-hour and 24-hour scenarios provide alternative valuations of \$100 and \$31 per unserved $kWh$, respectively. Table 12 summarizes these interruption-cost estimates.

## IX.I Economic Costs and Urban-Structure Heterogeneity

The baseline estimate in Table 5 implies an average reduction of approximately 2.8 percent in non-residential electricity consumption following a natural disaster. Because the outcome measures the log change in electricity consumption relative to the month preceding the disaster, I use each treated MSA's preceding-month consumption as the baseline quantity. For each treated MSA-event $(c, m, y)$, the implied reduction in electricity consumption is

$$\widehat{LostLoad}_{c,m,y} = Load_{c,m-1,y} \left[1 - \exp\left(\widehat{\tau}^{ATE}\right)\right] \tag{14}$$

[16]The ICE Calculator reports several normalized measures of interruption costs. Cost per $kW$ normalizes the average interruption cost by the customer's average electricity demand, while cost per unserved $kWh$ additionally accounts for interruption duration by normalizing the cost by the quantity of electricity that would have been consumed during the interruption. For a given interruption duration, unserved electricity is approximately equal to average demand in $kW$ multiplied by the duration in hours, so cost per unserved $kWh$ can be approximated by dividing cost per $kW$ by interruption duration.

Table 12: Non-Residential Electricity Interruption Cost Estimates

| Duration of Power Interruption Event | Cost per Event | Cost per kW | Cost per Unserved kWh | Cost per CMI |
|---|---|---|---|---|
| Momentary | $548 | $30 | $358 | $110 |
| 2 Hours | $3,571 | $200 | $100 | $30 |
| 8 Hours | $7,708 | $418 | $52 | $16 |
| 24 Hours | $13,878 | $747 | $31 | $10 |

*Notes:* This table reports non-residential electricity interruption cost estimates from the Interruption Cost Estimate (ICE) Calculator report. Cost per event measures the estimated economic cost incurred by an average non-residential customer for a single interruption event of the specified duration.
Cost per kW expresses the interruption cost relative to the customer's electricity demand, measured in kilowatts. Cost per unserved kWh measures the economic cost associated with one kilowatt-hour of electricity that would have been consumed but is not supplied because of the interruption. Cost per CMI expresses the interruption cost per customer minute interrupted, where customer minutes interrupted (CMI) are calculated as the number of affected customers multiplied by the duration of the interruption in minutes. The economic-loss calculations in this paper use the cost per unserved kWh for the 2-hour, 8-hour, and 24-hour interruption scenarios, corresponding to $100, $52, and $31 per unserved kWh, respectively.

I then translate this reduction into monetary losses using the ICE cost per unserved $kWh$. For outage-duration scenario $d$,

$$\widehat{EconomicLoss}^{,d}_{c,m,y} = 1000 \times \widehat{LostLoad}_{c,m,y} \times ICE^d \tag{15}$$

where the factor 1000 converts $MWh$ to $kWh$ and $ICE^d$ denotes the corresponding non-residential interruption cost per unserved $kWh$. I calculate these implied losses for each treated MSA-event and summarize their distribution across the 35 treated events.

Table 13 translates the baseline estimated reduction in non-residential electricity consumption into implied economic losses and reports their magnitude relative to monthly MSA GDP. Panel A first shows the corresponding reduction in electricity use. For the median treated MSA-event, preceding-month non-residential electricity consumption is approximately 216.1 $GWh$, implying a reduction of about 6.1 $GWh$.

Panel B translates these reductions into monetary losses under the alternative ICE scenarios, while Panel C expresses the resulting losses relative to monthly MSA GDP. Under the baseline 8-hour valuation, the median implied economic loss is approximately $318 million per treated MSA-event, equivalent to 13.6 percent of monthly MSA GDP. I focus on the median because the distribution of implied losses is right-skewed. Thus, the baseline 2.8 percent reduction in non-residential electricity consumption corresponds to an economic loss equivalent to approximately 14 percent of monthly local economic activity for the median treated MSA-event.

Table 13: Estimated Electricity Reductions and Economic Costs of Natural Disaster-Induced Electricity Disruptions

| | | | |
|---|---|---|---|
| *Panel A: Implied Reduction in Electricity Consumption per Treated MSA-Event* | | | |
| | *Preceding-Month Electricity Load* | | |
| Mean (GWh) | 343.640 | | |
| Median (GWh) | 216.136 | | |
| 25th percentile (GWh) | 81.893 | | |
| 75th percentile (GWh) | 585.524 | | |
| | *Implied Reduction in Electricity Consumption* | | |
| Mean (GWh) | 9.735 | | |
| Median (GWh) | 6.123 | | |
| 25th percentile (GWh) | 2.320 | | |
| 75th percentile (GWh) | 16.587 | | |
| *Panel B: Economic Loss per Treated MSA-Event* | | | |
| | Interruption Cost Estimate (ICE) Outage-Duration Scenario | | |
| | 2 Hours | 8 Hours | 24 Hours |
| Mean ($ million) | 974 | 506 | 302 |
| Median ($ million) | 612 | 318 | 190 |
| 25th percentile ($ million) | 232 | 121 | 72 |
| 75th percentile ($ million) | 1,660 | 863 | 514 |
| *Panel C: Economic Loss Relative to Monthly MSA GDP* | | | |
| | 2 Hours | 8 Hours | 24 Hours |
| Mean (%) | 31.98 | 16.63 | 9.92 |
| Median (%) | 26.23 | 13.64 | 8.13 |
| 25th percentile (%) | 14.76 | 7.67 | 4.57 |
| 75th percentile (%) | 52.61 | 27.35 | 16.31 |
| Treated MSA-events | | 35 | |

*Notes:* This table reports the implied reductions in electricity consumption and corresponding economic costs of disaster-induced electricity disruptions for treated MSA-events. The estimated electricity reduction is based on the contemporaneous average treatment effect reported in Table 5, $\hat{\beta} = -0.0287$.
For each treated MSA-event, counterfactual electricity consumption is approximated using electricity load in the preceding month, and the implied reduction in electricity consumption is calculated as $\text{Load}_{m,t-1}\left(1 - \exp(\hat{\beta})\right)$.
Panel A reports the distribution of preceding-month electricity load and the corresponding implied reductions in electricity consumption across treated MSA-events. Electricity quantities are reported in gigawatt-hours (GWh).
Panel B converts the implied electricity reductions into economic losses by applying interruption-cost estimates from the Interruption Cost Estimate (ICE) Calculator under alternative outage-duration scenarios of 2, 8, and 24 hours.
The corresponding interruption costs are $100, $52, and $31 per kWh of estimated unserved electricity, respectively.
Panel C expresses the resulting economic losses as a percentage of approximate monthly MSA GDP.
Monthly MSA GDP is constructed by aggregating annual county-level GDP from the Bureau of Economic Analysis to the MSA level and dividing annual MSA GDP by 12. The sample contains 35 treated MSA-events for which electricity-load and GDP data can be matched.

I next translate the heterogeneous treatment effects across the three dimensions of urban structure into implied economic losses. Table 6 shows that, relative to the baseline median loss equivalent to 13.6% of monthly MSA GDP, a one-standard-deviation increase in industrial concentration raises the implied loss to 31.8%. In contrast, a one-standard-deviation increase in road-network centralization or intra-city goods sourcing reduces the corresponding loss to 4.5% and 7.6% of monthly MSA GDP, respectively. These magnitudes illustrate the substantial economic implications of the heterogeneity in disaster responses across urban structures.

Table 14: Economic Implications of Urban Structure Intensity

| | Baseline (Mean) | Industrial Composition | Road Connectivity | Intra-city Goods Sourcing |
|---|---|---|---|---|
| | | +1 Standard Deviation | | |
| *Panel A: Marginal Treatment Effect and Implied Electricity Loss* | | | | |
| Marginal treatment effect | −0.0287 | −0.0684 | −0.0094 | −0.0159 |
| Implied electricity loss (%) | 2.83 | 6.61 | 0.94 | 1.58 |
| Change in implied loss relative to baseline (%) | — | +133.3 | −66.9 | −44.2 |
| *Panel B: Economic Loss under the 8-Hour ICE Scenario* | | | | |
| Mean ($ million) | 506 | 1,180 | 167 | 283 |
| Median ($ million) | 318 | 742 | 105 | 178 |
| 25th percentile ($ million) | 121 | 282 | 40 | 68 |
| 75th percentile ($ million) | 863 | 2,013 | 285 | 482 |
| *Panel C: Economic Loss Relative to Monthly MSA GDP* | | | | |
| Mean (%) | 16.63 | 38.80 | 5.50 | 9.29 |
| Median (%) | 13.64 | 31.82 | 4.51 | 7.62 |
| 25th percentile (%) | 7.67 | 17.90 | 2.54 | 4.28 |
| 75th percentile (%) | 27.35 | 63.82 | 9.04 | 15.28 |
| Treated MSA-events | 35 | 35 | 35 | 35 |

*Notes:* This table reports how the estimated economic costs of disaster-induced electricity disruptions vary with urban-structure intensity. The Baseline column evaluates the treatment effect with all urban-structure characteristics and population density fixed at their sample means. Each of the remaining columns evaluates the treatment effect after increasing the indicated urban-structure measure by one standard deviation above its sample mean, while holding the other urban-structure characteristics and population density at their sample means. The corresponding marginal treatment effects are taken from Table 6.
Implied electricity loss is calculated as $100 \times [1 - \exp(\widehat{\tau}_s)]$, where $\widehat{\tau}_s$ denotes the marginal treatment effect under the corresponding urban-structure scenario. The change relative to baseline reports the percentage change in the resulting implied electricity loss relative to the baseline case.
Economic losses are valued using the 8-hour Interruption Cost Estimate (ICE) scenario of $52 per unserved kWh. For each treated MSA-event, the implied reduction in electricity consumption is calculated using preceding-month electricity load. Panel B reports the distribution of implied economic losses across treated MSA-events, while Panel C expresses these losses as a percentage of approximate monthly MSA GDP. Monthly MSA GDP is constructed by aggregating annual county-level GDP from the Bureau of Economic Analysis to the MSA level and dividing annual MSA GDP by 12. The sample contains 35 treated MSA-events for which electricity-load and GDP data can be matched.

## IX.II Economic Implications of Health Industry Concentration

The industry-specific results from SectionVI reveal that greater concentration in Health and Education (HE) substantially amplifies the decline in electricity consumption following a natural disaster.

This finding is particularly relevant for the economic valuation of these disruptions because the ICE model also indicates higher interruption costs only for healthcare establishments. This provides an additional margin through which greater HE concentration may increase the economic consequences of disaster-induced electricity disruptions, beyond its larger estimated reduction in electricity consumption. I therefore examine the economic implications of greater HE concentration by combining these two margins, first translating its larger electricity response using the common ICE valuation and then incorporating the healthcare-specific adjustment to interruption costs.

To preserve the baseline treatment effect as the common benchmark, I use the industry group–specific specification only to measure the incremental effect of greater HE concentration. Moving $HHI_{HE}$ from its mean to one standard deviation above its mean changes the estimated treatment effect from $-0.0079$ to $-0.0282$, an incremental effect of $-0.0203$ log points. Applying this change to the baseline treatment effect of $-0.0287$ yields a counterfactual effect of $-0.0491$, corresponding to a 4.79 percent reduction in non-residential electricity consumption, approximately 69 percent larger than under the baseline[17]. The second margin arises from differences in the economic cost of electricity interruptions across industries. The non-residential ICE model estimates approximately 37.3 percent higher positive interruption costs for healthcare establishments, conditional on other customer and interruption characteristics[18]. I therefore value the HE counterfactual using both the common \$52 per unserved $kWh$ valuation and the healthcare-adjusted \$71.40 valuation and report the resulting economic losses as a share of monthly MSA GDP[19].

Table 15 reports the economic implications of a one-standard-deviation increase in HE concentration, $HHI_{HE}$. Applying the healthcare-specific economic costs of \$71.40 per unserved $kWh$ increases the median implied economic loss under greater HE concentration from approximately \$538 million to \$739 million, raising the corresponding loss from 23.05 percent to 31.66 percent of monthly MSA GDP. This additional increase illustrates how the higher interruption costs associated

[17]More specifically, I construct the baseline-adjusted HE counterfactual as

$$\begin{aligned}\widehat{\tau}_{HE}^{adj} &= \widehat{\tau}^{base} + \left(\widehat{\tau}_{HE,\mu+\sigma}^{sub} - \widehat{\tau}_{HE,\mu}^{sub}\right) \\ &= -0.0287 + [-0.0282 - (-0.0079)] = -0.0491 \qquad (16)\end{aligned}$$

[18]The non-residential ICE model uses LASSO to select industry indicators entering the interruption-cost model. Healthcare is retained in the generalized linear model (GLM) for positive interruption costs, with an estimated coefficient of 0.3173. Because the GLM uses a log link, the coefficient implies a multiplicative difference of $\exp(0.3173) \simeq 1.373$ in positive interruption costs. Applying this adjustment to the common 8-hour ICE valuation gives $ICE_{Health} = 52 \times \exp(0.3173) \simeq \$71.40$ per unserved $kWh$.

[19]Because $HHI_{HE}$ combines Health and Education whereas the ICE adjustment applies specifically to healthcare establishments, applying the healthcare-specific valuation to the entire HE counterfactual also assigns the higher healthcare interruption cost to the education component.

with healthcare activity can further amplify the economic consequences of greater HE concentration beyond its larger disaster-induced reduction in electricity consumption. From a resilience perspective, these findings suggest that maintaining electricity service and prioritizing rapid restoration may be particularly important in HE-concentrated cities. These cities exhibit greater vulnerability to natural disasters, while interruptions to healthcare activity can impose relatively high economic costs.

Table 15: Economic Implications of Health and Education Industry Concentration

| | Baseline (Mean Urban Structure) | HE Concentration (+1 SD) | |
|---|---|---|---|
| *Panel A: Treatment Effect and Implied Electricity Loss* | | | |
| Baseline treatment effect | −0.0287 | −0.0287 | |
| Incremental HE concentration effect | — | −0.0203 | |
| Baseline-adjusted treatment effect | −0.0287 | −0.0491 | |
| Implied electricity loss (%) | 2.83 | 4.79 | |
| Change in implied electricity loss relative to baseline (%) | — | 69.0 | |
| *Panel B: Economic Valuation* | | | |
| | Average ICE Valuation | Average ICE Valuation | Healthcare-Specific ICE Valuation |
| ICE cost per unserved kWh | $52.00 | $52.00 | $71.40 |
| Mean economic loss ($ million) | 506 | 855 | 1,174 |
| Median economic loss ($ million) | 318 | 538 | 739 |
| 25th percentile ($ million) | 121 | 204 | 280 |
| 75th percentile ($ million) | 863 | 1,458 | 2,002 |
| *Panel C: Economic Loss Relative to Monthly MSA GDP* | | | |
| Mean (%) | 16.63 | 28.11 | 38.59 |
| Median (%) | 13.64 | 23.05 | 31.66 |
| 25th percentile (%) | 7.67 | 12.97 | 17.81 |
| 75th percentile (%) | 27.35 | 46.23 | 63.47 |
| Treated MSA-events | 35 | 35 | 35 |

*Notes:* This table reports the economic implications of a one-standard-deviation increase in Health and Education (HE) concentration relative to the baseline.
Panel A reports the estimated treatment effects and corresponding implied reductions in non-residential electricity consumption. Panel B reports the implied economic losses under the common 8-hour non-residential ICE valuation of $52 per unserved $kWh$ and the healthcare-adjusted valuation of $71.40 per unserved $kWh$. The healthcare-adjusted valuation applies the higher interruption cost estimated for healthcare establishments to the HE counterfactual, which also includes Education. Panel C expresses the resulting economic losses as a percentage of monthly MSA GDP. The sample contains 35 treated MSA-events with matched electricity-load and GDP data.

# X Conclusion

Urban agglomeration has long been recognized as a fundamental source of productivity and economic growth, yet much less is known about how the economic structures underlying these benefits shape cities' resilience to natural disasters[20]. This paper shows that natural disasters generate substantial short-run economic disruptions and that their magnitude varies systematically with urban economic structure. Greater industrial concentration amplifies disaster-induced losses, whereas more centralized road networks and stronger intra-city goods sourcing mitigate them. Moreover, the consequences of industrial concentration depend on where that concentration occurs, with Accommodation, Wholesale Trade, and Transportation and Health and Education associated with greater vulnerability.

The broader lesson is that understanding the economic consequences of natural disasters requires looking beyond average effects and examining the sources of heterogeneity in local resilience. For instance, the greater vulnerability associated with Accommodation, Wholesale Trade, and Transportation is consistent with the prominent role of these activities in individuals' everyday mobility patterns, while the greater vulnerability associated with Health and Education is particularly consequential given the high economic costs of electricity interruptions for healthcare activities. These patterns suggest that resilience efforts may benefit from preserving access to activities that are central to everyday mobility and maintaining reliable electricity service for activities with particularly high interruption costs. As natural disasters become more frequent and severe, accounting for such heterogeneity can help identify where resilience investments may have the greatest economic value and provide a more complete assessment of the benefits and vulnerabilities associated with urban agglomeration.

[20]For instance, Lin et al. (2024) show that, despite growing awareness of sea-level rise, people continued to move into coastal U.S. cities between 1990 and 2010 to capture the economic benefits of urban proximity. This illustrates the willingness to accept greater disaster exposure in exchange for economic opportunities.

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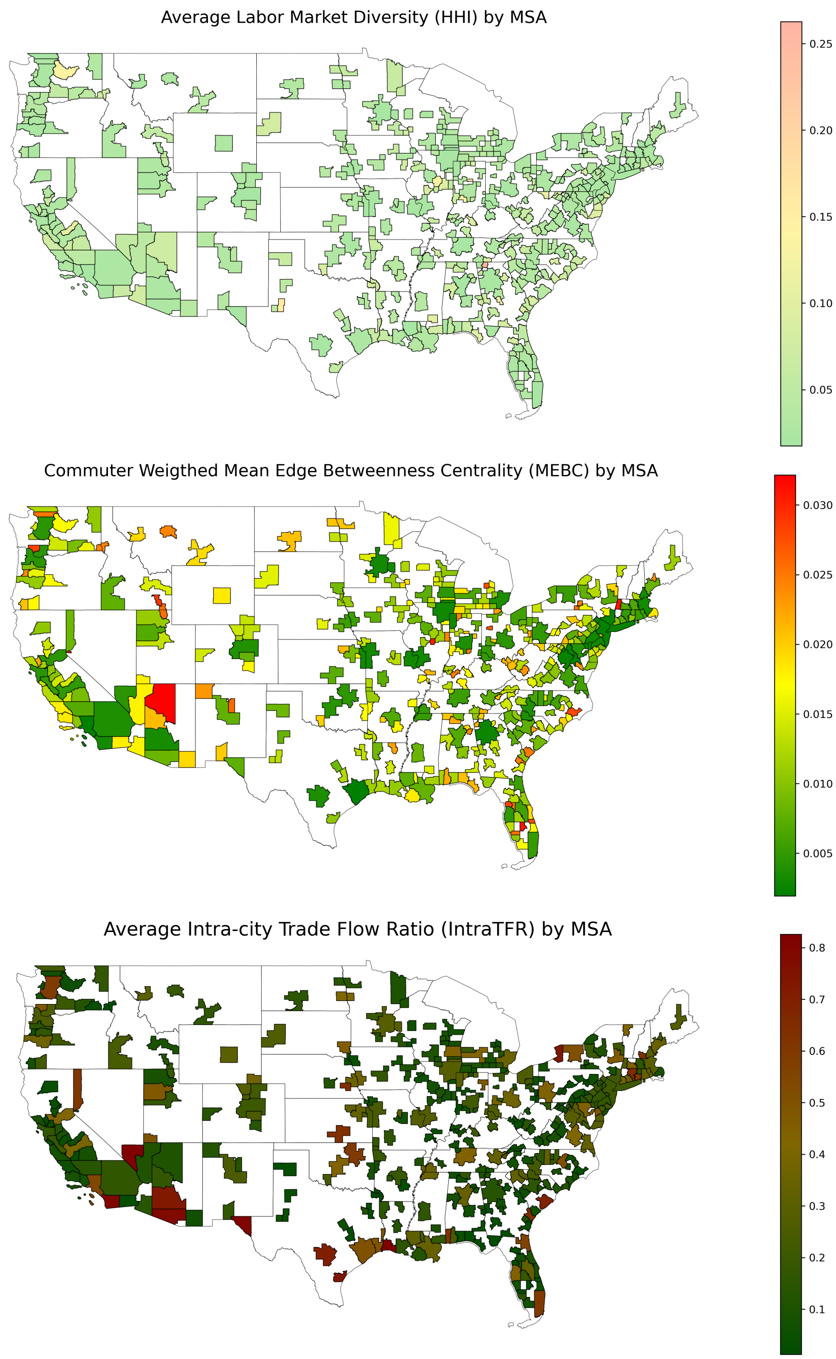
# XI Appendix

Figure 14: City-Level Concentration Measures (2017)




*Note*: This figure presents average concentration measures for the labor market, road network, and trade perspectives across MSAs from 2017.

### XI.I Treatment Effects across Alternative Control Specifications

The preferred specification in Section IV.II controls for time-varying determinants of electricity demand in both the event month and the month immediately preceding the event. This appendix examines how the estimated treatment effect varies with alternative control specifications.

The inclusion of conditions in both months reflects two features of the empirical setting. First, the outcome measures the change in electricity consumption between the event month and the preceding month, so weather-sensitive demand in either month contributes to the observed change. Second, natural-disaster exposure is not evenly distributed across the calendar year. As shown in Table 1, high-wind events occur throughout the year but are more concentrated in particular months. Because heating and cooling demand also varies systematically over the year, the timing of disaster exposure may coincide with weather-driven differences in electricity demand between the event month and the preceding month.

Table 16 examines the relevance of this control structure by sequentially introducing the time-varying controls used in the preferred specification. Column (1) provides an uncontrolled benchmark. Column (2) introduces event-month weather conditions, while Column (3) additionally accounts for weather conditions in the preceding month. Column (4) adds electricity-outage controls for both months, and Column (5) further includes the pre-1980 housing share, corresponding to the preferred specification. The estimated treatment effect remains negative and statistically significant across all specifications, although its magnitude varies with the set of controls included. Relative to the preferred specification, omitting all time-varying controls produces an estimated decline that is 0.17 percentage points larger. This difference is consistent with the uncontrolled specification reflecting not only changes associated with disaster exposure but also other time-varying determinants of electricity consumption that are accounted for in the preferred specification.

Comparing the specifications with event-month weather only and with weather conditions in both months illustrates the importance of accounting for preceding-month weather. The estimated treatment effect changes from $-0.0308$ to $-0.0287$ once preceding-month HDD and CDD are included, indicating that weather-sensitive electricity demand in the comparison month is relevant for estimating the disaster-related change in electricity consumption. Subsequent additions of the outage controls and the pre-1980 housing share produce comparatively small changes.

Table 16: Treatment Effects across Alternative Control Specifications

| | (1) No Controls | (2) Event-Month Weather | (3) Weather in Both Months | (4) Weather & Outages | (5) Preferred Specification |
|---|---|---|---|---|---|
| *Panel A: Marginal Treatment Effect* | | | | | |
| Marginal treatment effect | $-0.0305^{***}$ (0.0056) | $-0.0308^{***}$ (0.0055) | $-0.0287^{***}$ (0.0056) | $-0.0285^{***}$ (0.0055) | $-0.0287^{***}$ (0.0056) |
| Difference from preferred specification | $-0.0017$ | $-0.0021$ | 0.0000 | 0.0002 | |
| *Panel B: Weather Controls* | | | | | |
| Event-month HDD | — | $0.0010^{***}$ (0.0003) | $0.0098^{***}$ (0.0018) | $0.0098^{***}$ (0.0018) | $0.0098^{***}$ (0.0018) |
| Event-month CDD | — | $0.0031^{***}$ (0.0008) | $0.0201^{***}$ (0.0035) | $0.0201^{***}$ (0.0035) | $0.0201^{***}$ (0.0035) |
| Preceding-month HDD | — | — | $-0.0093^{***}$ (0.0017) | $-0.0093^{***}$ (0.0017) | $-0.0093^{***}$ (0.0017) |
| Preceding-month CDD | — | — | $-0.0192^{***}$ (0.0033) | $-0.0192^{***}$ (0.0033) | $-0.0193^{***}$ (0.0033) |
| Event- and preceding-month outages | No | No | No | Yes | Yes |
| Pre-1980 housing share | No | No | No | No | Yes |
| Observations | | | 30,221 | | |

*Notes:* The table reports the event-month ($h = 0$) treatment effect under alternative control specifications while holding the estimation sample fixed. The marginal treatment effect is evaluated at the sample means of the urban-structure characteristics. Column (1) omits time-varying controls. Column (2) includes event-month heating and cooling degree days. Column (3) additionally includes heating and cooling degree days in the preceding month. Column (4) adds electricity-outage controls for both the event and preceding months. Column (5) additionally controls for the share of the housing stock built before 1980 and corresponds to the preferred specification used in the main analysis. All specifications include the same treatment interactions with industrial composition, road connectivity, intra-city goods sourcing, and population density, as well as event-month and preceding-month time fixed effects. Standard errors are clustered at the MSA level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

## XI.II Building Vintage and Disaster Vulnerability

Building vintage may influence electricity consumption through both differences in energy use and differences in vulnerability to natural disasters. Older and newer buildings differ in insulation, heating and cooling technologies, energy efficiency, and other characteristics that can affect electricity demand. Building age may also be associated with differences in structural characteristics and construction standards that affect susceptibility to disaster-related damage. While the main specification controls for the share of buildings constructed before 1980, it does not allow the effect of disaster exposure itself to vary with building vintage. I therefore augment the main specification by interacting disaster exposure with the share of pre-1980 buildings:

$$\Delta D_{c,m,y} \times BuildingSh_{\text{Pre-1980},c,y} \tag{17}$$

to Equation 9, while retaining the level control for the pre-1980 building share and all other controls in the baseline specification.

Table 17 compares the estimated heterogeneity coefficients from the baseline specification with those from the augmented specification. The estimated relationships between industrial concentration, road centrality, intra-city goods sourcing, and disaster resilience remain similar in magnitude and statistical significance after allowing the disaster effect to vary with building vintage. The main heterogeneity results therefore do not appear to be driven by differences in the age composition of the local building stock.

The interaction between disaster exposure and the share of buildings constructed before 1980 in Table 17 is positive and statistically significant, indicating that MSAs with a larger share of pre-1980 buildings experience a smaller decline in electricity consumption from the preceding non-disaster month to the disaster month. One possible explanation is that building vintage also captures persistent differences in urban form, as durable building stocks shape the evolution of cities over time (Siodla, 2015). In particular, older building stocks may be associated with more compact urban development, which can provide greater connectivity and accessibility under disruptions (Alawadi et al., 2026). The mitigating association with a larger pre-1980 building share may therefore partly reflect these differences in urban form, in addition to differences in the physical characteristics of the building stock.

Table 17: Robustness to Building-Vintage Heterogeneity

| | | Change in Monthly MSA Electricity Load | |
|---|---|---|---|
| | | Baseline | Building-Vintage Interaction |
| | | (1) | (2) |
| *Panel A: Average Treatment Effect* | | | |
| | Average treatment effect | $-0.0287^{***}$ (0.0056) | $-0.0233^{***}$ (0.0057) |
| *Panel B: Interaction Coefficients* | | | |
| Labor Market | Industrial Composition ($HHI$) | $-1.585^{***}$ (0.288) | $-1.471^{***}$ (0.254) |
| Road Network | Road Connectivity ($wMEBC$) | $3.049^{***}$ (0.939) | $3.327^{***}$ (0.846) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.0745^{***}$ (0.0275) | $0.0793^{***}$ (0.0269) |
| Demographics | Population Density | $-0.000422^{***}$ (0.0000872) | $-0.000397^{***}$ (0.0000822) |
| Building Stock | Pre-1980 Building Share | — | $0.0795^{***}$ (0.0227) |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics, population density, and, where applicable, the pre-1980 building share. Panel B reports the estimated interaction coefficients between natural disaster exposure and the corresponding characteristics. Column (1) reports the baseline specification. Column (2) additionally interacts natural disaster exposure with the share of the housing stock built before 1980. Both specifications control for heating degree days (HDD), cooling degree days (CDD), and electricity outages in both the event month and the preceding month, as well as the level of the pre-1980 building share. Event-month and preceding-month time fixed effects are included. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. Standard errors, clustered at the MSA level, are reported in parentheses. MSA-level electricity load is constructed using population-based weights. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

## XI.III Heterogeneous Employment Responses Across Urban Economic Structures

Table 18: Heterogeneous Treatment Effects by Urban Structure Intensity: Average Three-Month Employment Changes (CES-SAE)

| | | Treatment Effects Across Urban Structure Levels: Event Month ($h = 0$) | | | |
|---|---|---|---|---|---|
| | | Average Three-Month Employment Changes | | | Baseline Effect |
| Category | Variable | Low | High | Difference | |
| Labor Market | Industrial Composition ($HHI$) | $-0.2038$ (0.2872) | $-0.9451^{**}$ (0.4143) | $-0.7413$ (0.5614) | $-0.5744^{***}$ (0.2197) |
| Road Network | Road Connectivity ($wMEBC$) | $-0.8773^{***}$ (0.2900) | $-0.2715$ (0.2063) | $0.6059^{**}$ (0.2453) | |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $-0.8446^{***}$ (0.2770) | $-0.3042$ (0.1977) | $0.5404^{***}$ (0.1963) | |
| Observations: 30,436 | | | | | |

Note: Low and High columns report estimated treatment effects on average three-month employment changes (CES-SAE), evaluated at one standard deviation below and above the mean of each urban structure measure. The Difference column reports the difference between the High and Low treatment effects, while the Baseline column reports the treatment effect evaluated at the sample means. Because employment is measured in thousands of workers, the baseline estimate of (-0.5774) indicates that an ND-HIT month reduces the change in the average three-month employment count by approximately 0.58 thousand workers, or about 577 workers, relative to the change that would have occurred without an ND-HIT. Standard errors are clustered at the MSA level. Observations: 30,436. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

## XI.IV Average Operating Intensity

I replace monthly electricity consumption with its three-month moving average, which captures firms' average operating intensity over a broader time window while reducing the influence of transitory month-to-month fluctuations. This smoothing, however, comes at the cost of attenuating short-run variation, making the measure less sensitive to the immediate economic disruption caused by a natural disaster.

I first examine firms' operating responses over a broader temporal window. Whereas the baseline outcome captures the immediate response of commercial and industrial electricity consumption in the disaster month, the three-month moving average captures the average intensity of business operations over recent months. Comparing the two measures therefore provides a useful distinction between the immediate disruption associated with a natural disaster and its effect on firms' average operating intensity over a broader period.

Table 19 reports the event-month estimates using the logarithm of three-month average commercial and industrial electricity consumption as the outcome variable, which captures firms' average operating intensity over a broader temporal window. Compared with the baseline estimate of firms' immediate operating response, measured using the logarithm of event-month commercial and industrial electricity consumption and reported in Table 5, the estimated marginal effect of natural disaster exposure declines by approximately half, to 1.34 percent. This smaller effect is consistent with the three-month average smoothing the immediate disruption observed in the event month, resulting in a more moderate decline in firms' average operating intensity than in their contemporaneous operating intensity. Furthermore, the heterogeneous effects across urban economic structures differ from those obtained using monthly electricity consumption. Most notably, industrial concentration continues to play a significant role in shaping firms' average operating intensity following natural disasters. By contrast, the buffering effects of centralized road connectivity and intra-city goods sourcing become substantially weaker and are no longer statistically significant. These results suggest that road networks and local supplier linkages primarily mitigate immediate operational disruptions rather than shaping firms' average operating intensity following natural disasters. Industrial composition, however, remains an important determinant of firms' average operating intensity following natural disasters, even when operating activity is measured using a three-month average rather than monthly electricity consumption.

Table 19: Contemporaneous Effects of Natural Disaster Exposure on the Three-Month Average Commercial and Industrial Electricity Consumption

| | | Three-Month Average Log Electricity Consumption |
|---|---|---|
| | | Event Month ($h = 0$) |
| *Panel A: Average Treatment Effect* | | |
| | Average treatment effect | $-0.0134^{***}$ (0.00456) |
| *Panel B: Interaction Coefficients* | | |
| Labor Market | Industrial Composition ($HHI$) | $-0.741^{***}$ (0.235) |
| Road Network | Road Connectivity ($wMEBC$) | 0.768 (0.495) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | 0.00858 (0.0269) |
| Demographics | Population Density | $-0.000208^{**}$ (0.0000837) |
| | Observations | 25,263 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in three-month average commercial and industrial electricity consumption following natural disaster exposure, whereas negative coefficients indicate that they amplify it.

The dependent variable is the logarithm of the three-month average commercial and industrial electricity consumption,

$$\ln\left(\frac{Load_{it} + Load_{i,t-1} + Load_{i,t-2}}{3}\right),$$

where $Load_{it}$ denotes commercial and industrial electricity consumption in MSA $i$ during month $t$. Relative to the baseline specification based on monthly electricity consumption, this measure captures firms' average operating intensity while reducing the influence of transitory month-to-month fluctuations.

Treatment is defined by an eligible storm onset in the event month. The estimated coefficient of $-0.0134$ implies that, evaluated at the sample means of all urban-structure characteristics, natural disaster exposure reduces the three-month average commercial and industrial electricity consumption by approximately 1.34 percent relative to the corresponding counterfactual outcome.

Control MSAs are required to exhibit no treatment-status transitions during the relevant pre-event stabilization window. Standard errors, clustered at the MSA level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

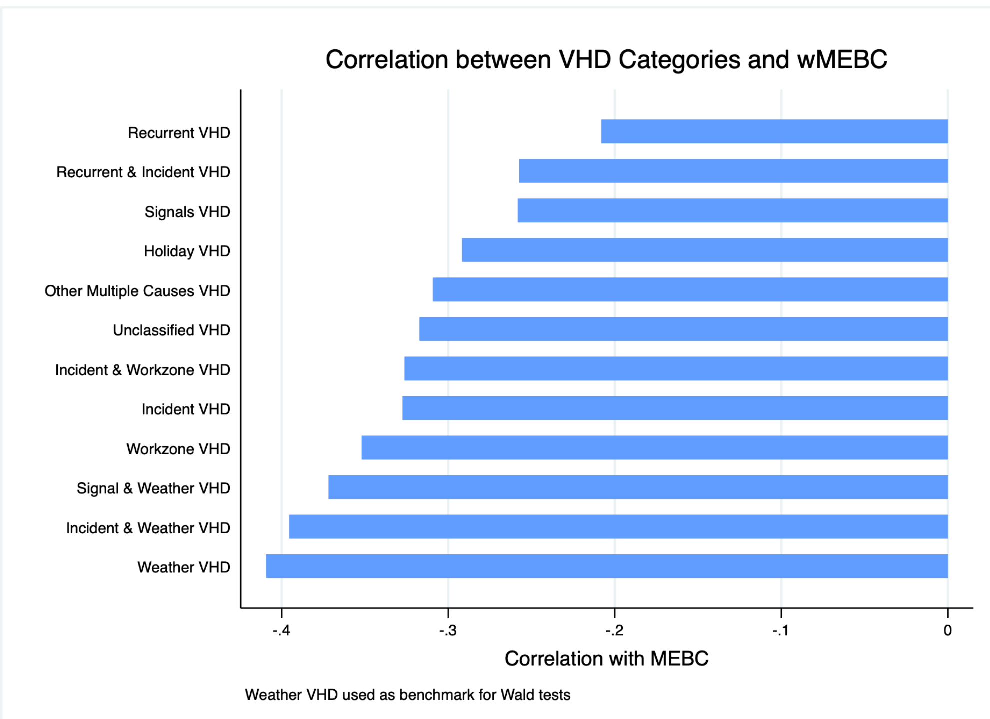
## XI.V Unadjusted Correlation Between Road Centrality and Traffic Congestion

Table 20: Standardized Unadjusted Correlations Between Weighted Mean Edge Between Centrality and Vehicle Hours of Delay

| Panel A. Overall Congestion | | |
|---|---|---|
| Measure | Standardized Correlation | |
| Total VHD | −0.263 | |
| **Panel B. VHD Categories** | | |
| VHD Category | Standardized Correlation | Difference from Weather |
| Weather VHD | −0.409 | Reference |
| Incident & Weather VHD | −0.395 | |
| Signal & Weather VHD | −0.372 | |
| Workzone VHD | −0.352 | * |
| Incident VHD | −0.327 | ** |
| Incident & Workzone VHD | −0.326 | ** |
| Unclassified VHD | −0.317 | *** |
| Other Multiple Causes VHD | −0.309 | *** |
| Holiday VHD | −0.292 | *** |
| Signals VHD | −0.258 | *** |
| Recurrent & Incident VHD | −0.257 | *** |
| Recurrent VHD | −0.208 | *** |

Notes: Entries report standardized correlations between weighted Mean Effective Betweenness Centrality (wMEBC) and Vehicle Hours of Delay (VHD), estimated from a covariance structural equation model (SEM), without controlling for any population or city size. Panel A reports the correlation between wMEBC and total VHD. Panel B decomposes total VHD into mutually exclusive delay categories. Weather VHD serves as the reference category. The "Difference from Weather" column reports the significance of Wald tests comparing each VHD–wMEBC correlation with the Weather VHD–wMEBC correlation. Blank entries indicate that the corresponding correlation is not statistically different from the Weather VHD correlation. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 15: Unadjusted correlations between VHD categories and $wMEBC$ under alternative adjustment specifications



## XI.VI Decomposing Industry Group–Specific $HHI$ Measures

While the industry group–specific $HHI$, $HHI_{gc}$, are designed to capture concentration associated with each broad industry group, they incorporate two distinct features of local industry structure simultaneously: the relative scale of the industry group in the local economy and the degree to which employment within that group is specialized in a small number of subsectors. Because $HHI_{gc}$ is constructed using total MSA employment as the denominator, variation in the measure can arise from either of these two dimensions. Accordingly, each group-specific HHI jointly captures the following two components.

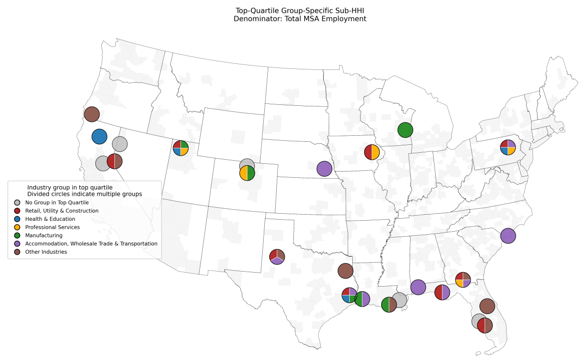
$$HHI_{gc} = \sum_{i \in g} \left( \frac{Emp_{ic}}{TotalEmp_c} \right)^2, \qquad g \in \{\text{RUC}, \text{HE}, \text{PS}, \text{Manu}, \text{ATT}, \text{Other}\}$$

$$\begin{aligned}
&= \sum_{i \in g} \left( \left( \frac{Emp_{ic}}{Emp_{gc}} \right) \times \left( \frac{Emp_{gc}}{TotalEmp_c} \right) \right)^2 \\
&= \left( \frac{Emp_{gc}}{TotalEmp_c} \right)^2 \sum_{i \in g} \left( \frac{Emp_{ic}}{Emp_{gc}} \right)^2 \\
&= Share_{gc}^2 \times WithinHHI_{gc} \qquad (18)
\end{aligned}$$

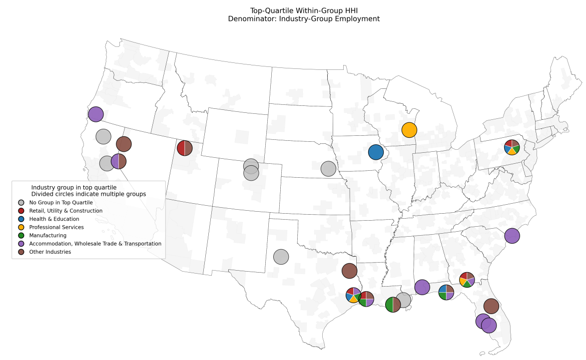
Equation 18 shows how $HHI_{gc}$ can be decomposed into two multiplicative components. The first component, $Share_{gc}^2$, is the squared employment share of industry group $g$ in MSA $c$, capturing the relative size of the sector within the local economy. The second component, $WithinHHI_{gc}$, measures the concentration of employment across industries within group $g$. Thus, the industry group–specific $HHI$, $HHI_{gc}$, jointly captures two dimensions of local industry structure, the scale of the industry group and the degree of within-group specialization. The scale dimension reflects the importance of industry group $g$ in the local economy, as measured by its share of total MSA employment. The specialization dimension reflects the extent to which employment within the broad industry group is concentrated in a small number of constituent industries rather than more evenly distributed across them. In other words, given that the overall employment concentration of an MSA ($HHI_c$) can be expressed as the sum of industry group–specific concentration measures ($HHI_{gc}$), and that each $HHI_{gc}$ is, by construction, the product of the industry's share of total city employment and its within-group concentration ($WithinHHI_{gc}$), each industry's contribution to aggregate employment concentration from SectionVI.III reflects both the group's economic importance in the local economy and the degree of concentration among its constituent industries.

Figure 16 compares the geographic distribution of treated cities classified in the top quartile under the $HHI_{gc}$ and $WithinHHI$ measures. By removing the industry-size component from the concentration measure, the figure illustrates how the spatial distribution of highly concentrated cities changes when concentration is defined solely by the distribution of employment across detailed industries within each broad industry group. Panel C highlights the overlap between the two measures, identifying cities that remain highly concentrated after accounting for industry-group size. Cities that appear only under the industry group–specific $HHI$ derive their high-concentration classification from the combination of industry-group scale and within-group specialization, whereas cities that remain highly concentrated under $WithinHHI$ exhibit substantial specialization within the industry group itself, independent of its relative size in the local economy. This decomposition allows me to investigate whether the observed heterogeneity across industry groups in shaping natural disaster impacts is driven primarily by differences in sectoral scale or by differences in within-group specialization.

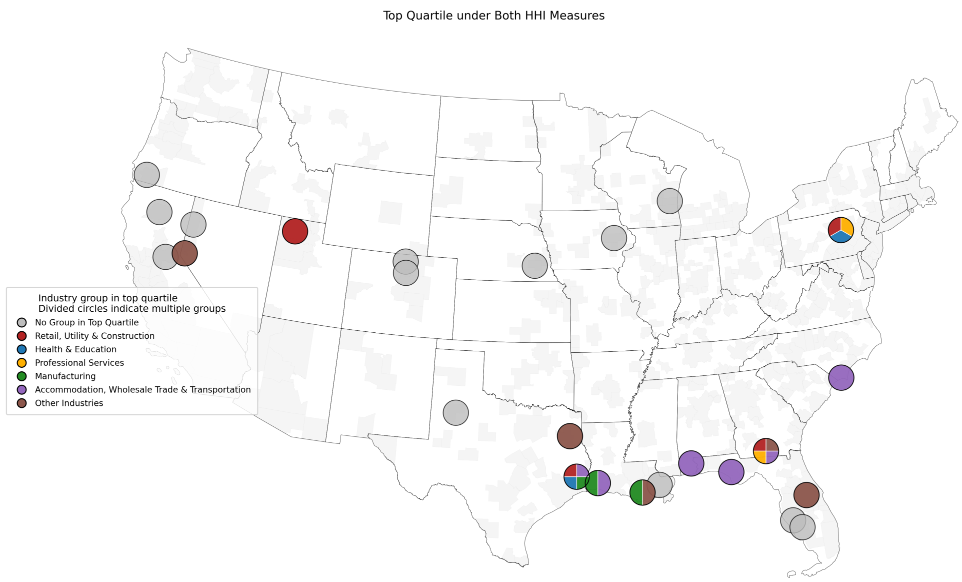
Figure 16: Geographic Comparison of High Concentration Across Different $HHI$ Measures




(a) Top-quartile Industry Group–Specific $HHI$




(b) Top-quartile $WithinHHI$




(c) Top quartile under both measures

*Note*: Colored circles indicate industry groups in the top quartile. Divided circles indicate multiple industry groups in the top quartile within the same MSA. Gray circles indicate that no industry group is in the top quartile.

### XI.VI.1 Results Using Within-Group $HHI$ Measures

Given that the heterogeneous mitigating and exacerbating effects of Manufacturing, ATT, and HE documented in Section VI.III may arise through two distinct dimensions, it is useful to distinguish whether these effects reflect the relative importance of each industry group in the local economy or the degree of specialization among the detailed industries within the group. To isolate the latter dimension, I next replace the industry group–specific $HHI_{gc}$ measures with $WithinHHI_{gc}$ measures, which capture the degree of employment concentration across detailed industries within each broad industry group independently of the group's employment share in the local economy. Comparing the results across the two measures allows me to assess whether the heterogeneous effects across industry groups are primarily associated with sectoral scale or with within-group specialization.

Table 21 reports the results using $WithinHHI$ measures, thereby isolating the role of employment concentration across detailed industries within each broad industry group. The results show that HE is the only industry group that exhibits a statistically significant exacerbating effect, indicating that greater concentration of employment in a relatively small number of HE sub-industries itself amplifies the economic consequences of natural disasters. Disaster vulnerability therefore increases not only when HE represents a larger share of metropolitan employment, but also when employment within the sector is concentrated in relatively few detailed industries. For example, an MSA whose HE employment is dominated by a small number of activities, such as hospitals or universities, exhibits greater within-group specialization than an MSA with a more diversified mix of hospitals, outpatient care, nursing facilities, schools, universities, social services, and other HE activities. In contrast, the exacerbating effect of ATT reported in Table 9 disappears. This finding suggests that the vulnerability associated with ATT primarily reflects its overall importance within the metropolitan economy rather than the distribution of employment across detailed ATT industries. Cities with a larger ATT workforce are therefore more vulnerable to disaster-related disruptions, regardless of how employment is allocated across the sector's constituent industries. Similarly, the mitigating effect previously observed for manufacturing with the industry group–specific $HHI$ measure is no longer statistically significant when using $WithinHHI$. This result indicates that the buffering effect of manufacturing likewise reflects the sector's overall importance within the local economy rather than the distribution of employment across detailed manufacturing industries. In other words, cities with a larger manufacturing workforce appear to be more resilient, irrespective of the sector's internal composition.

Table 22 and Figure 17 illustrate how the estimated marginal treatment effect varies with the level of each $WithinHHI$, capturing differences in the extent to which employment within each broad industry group is concentrated in a small number of detailed industries. Consistent with the interaction estimates reported in Table 21, meaningful heterogeneity is observed only for the HE sector. Specifically, increasing within-group concentration in the HE sector from the sample mean

Table 21: Marginal Effects in the Event Month ($h = 0$): Within-Industry-Group $HHI$ Measures

| | | Change in Monthly MSA Electricity Load |
|---|---|---|
| | | Event Month ($h = 0$) |
| *Panel A: Average Treatment Effect* | | |
| | Average treatment effect | $-0.0117$ (0.0084) |
| *Panel B: Interaction Coefficients* | | |
| Labor Market | Retail, Utilities, and Construction ($WithinHHI_{\text{RUC}}$) | $-0.0352$ (0.4027) |
| | Health and Education ($WithinHHI_{\text{HE}}$) | $-0.3025^{***}$ (0.0919) |
| | Professional Services ($WithinHHI_{\text{PS}}$) | 0.0058 (0.2688) |
| | Manufacturing ($WithinHHI_{\text{Manu}}$) | 0.0991 (0.1131) |
| | Accommodation, Transportation, and Trade ($WithinHHI_{\text{ATT}}$) | $-0.1560^{**}$ (0.0743) |
| | Other Industries ($WithinHHI_{\text{Other}}$) | 0.0491 (0.0782) |
| Road Network | Road Connectivity ($wMEBC$) | $3.670^{*}$ (1.981) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.0897^{*}$ (0.0526) |
| Demographics | Population Density | $-0.000188$ (0.000116) |
| | Observations | 30,221 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all within-industry-group concentration measures, the remaining urban-structure characteristics, and population density. Panel B reports the estimated interaction coefficients between natural disaster exposure and each within-industry-group concentration measure, together with the remaining urban-structure characteristics. The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. The within-industry-group concentration measures ($WithinHHI$) capture the degree of employment concentration across detailed industries within each broad industry group. Standard errors, clustered at the MSA level, are reported in parentheses. MSA-level electricity load is constructed using population-based weights. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

to one standard deviation above the mean increases the estimated decline in economic activities following a natural disaster from 1.98 percent to 4.79 percent, more than doubling the estimated treatment effect. For the remaining industry groups, the estimated treatment e ffects vary across the observed range of within-group concentration, but these differences are not statistically distinguishable from zero.

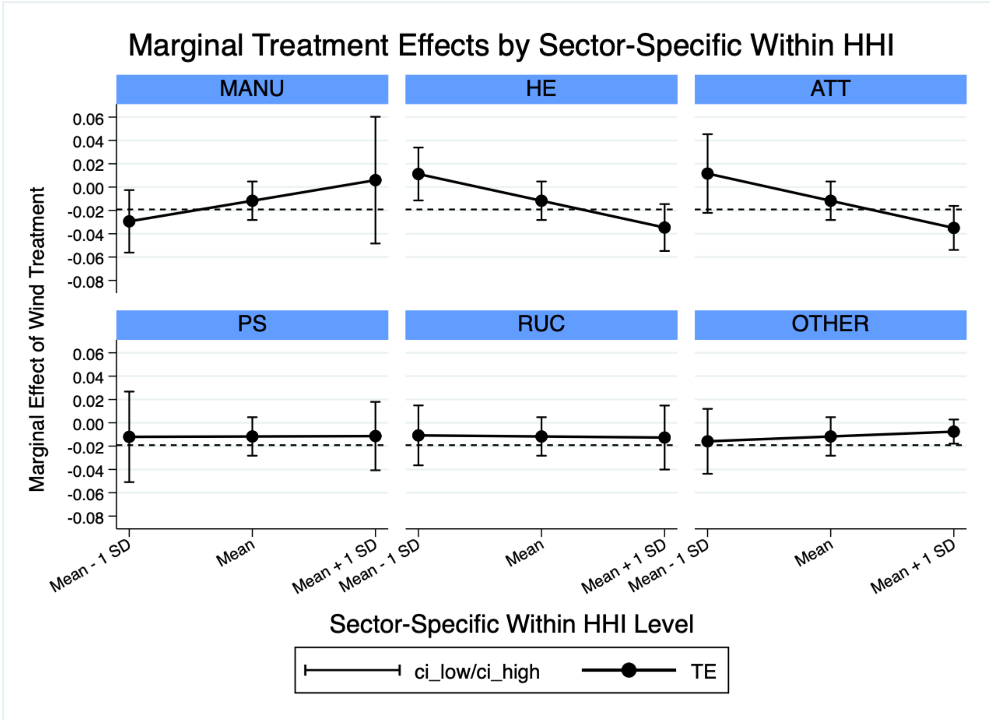
Overall, the mechanism through which aggregate industrial concentration exacerbates the impacts of natural disasters operates through the heterogeneous roles of broad industry groups. Specifically, the ATT and HE industry groups amplify disaster impacts, whereas the Manufacturing group mitigates them. However, the mechanisms underlying these heterogeneous effects differ across sectors. For the ATT and Manufacturing groups, the heterogeneous effects are primarily driven by the sectors' relative scale and economic importance within the local economy, rather than by the degree to which employment is concentrated in a small number of detailed industries. In contrast, the exacerbating effect of the HE group persists even after abstracting from the sector-size component. This finding indicates that employment concentration within a relatively small number of HE sub-industries independently amplifies the impacts of natural disasters. Consequently, the vulnerability associated with the HE sector cannot be attributed solely to its importance in the local economy. Rather, internal specialization within the sector constitutes an additional and distinct source of disaster vulnerability.

Table 22: Heterogeneous Treatment Effects by Within-Industry-Group Concentration

| | Treatment Effects Across Within-Industry-Group Concentration | | | Baseline Effect |
|---|---|---|---|---|
| | Monthly Electricity Load | | | |
| Variable | Low | High | Difference | |
| Manufacturing ($WithinHHI_{MANU}$) | $-0.0294^{**}$ (0.0137) | 0.0059 (0.0277) | 0.0353 (0.0403) | |
| Health & Education ($WithinHHI_{HE}$) | 0.0112 (0.0116) | $-0.0347^{***}$ (0.0103) | $-0.0459^{***}$ (0.0139) | |
| Accommodation, Wholesale Trade & Transportation ($WithinHHI_{ATT}$) | 0.0116 (0.0172) | $-0.0350^{***}$ (0.0096) | $-0.0466^{**}$ (0.0222) | $-0.0117$ (0.0084) |
| Professional Services ($WithinHHI_{PS}$) | $-0.0121$ (0.0198) | $-0.0114$ (0.0150) | 0.0007 (0.0308) | |
| Retail, Utilities, and Construction ($WithinHHI_{RUC}$) | $-0.0108$ (0.0131) | $-0.0126$ (0.0140) | $-0.0019$ (0.0213) | |
| Other Industries ($WithinHHI_{OTHER}$) | $-0.0159$ (0.0142) | $-0.0076$ (0.0053) | 0.0083 (0.0132) | |
| Observations: 30,221 | | | | |

*Notes:* The Low and High columns report treatment effects evaluated at one standard deviation below and above the sample mean of the corresponding within-industry-group concentration measure, respectively, while holding the remaining within-industry-group concentration measures, other urban-structure characteristics, and population density at their sample means. The Difference column reports the change in the estimated treatment effect from the Low to the High concentration level. The Baseline Effect column reports the treatment effect evaluated with all within-industry-group concentration measures, remaining urban-structure characteristics, and population density fixed at their sample means. The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Standard errors, clustered at the MSA level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 17: Marginal Treatment Effects by Industry Group–Specific




Estimated marginal treatment effects of episodic natural disasters evaluated at the mean and one standard deviation above and below the mean of each $withinHHI$, holding all other covariates at their sample means. Error bars denote 95% confidence intervals.

## XI.VII Robustness Checks

### XI.VII.1 Alternative Weighting Scheme for Electricity Consumption Distribution

In the baseline analysis, I allocate utility-level electricity load to MSAs using the population shares of the counties comprising each MSA to construct monthly commercial and industrial electricity consumption. As an alternative, I instead use CBP employment shares as the allocation weights. Figure 3 shows that the resulting MSA-level electricity consumption series are highly consistent across the two weighting schemes. Consistent with this finding, Table 23 shows that re-estimating the baseline specification using the employment-weighted electricity consumption measure yields results that are highly similar to the baseline estimates.

Table 23: Robustness to Alternative Electricity-Load Weighting Scheme

| | | Change in Monthly MSA Electricity Load |
|---|---|---|
| | | CBP Employment Weights |
| *Panel A: Average Treatment Effect* | | |
| | Average treatment effect | $-0.0284^{***}$ (0.0055) |
| *Panel B: Interaction Coefficients* | | |
| Labor Market | Industrial Composition ($HHI$) | $-1.634^{***}$ (0.295) |
| Road Network | Road Connectivity ($wMEBC$) | $2.941^{***}$ (0.932) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.0696^{***}$ (0.0268) |
| Demographics | Population Density | $-0.000432^{***}$ (0.0000854) |
| | Observations | 30,221 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of the urban-structure characteristics and population density. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. MSA-level electricity load is constructed using County Business Patterns (CBP) employment-based weights rather than the population-based weights used in the baseline specification.
The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the estimated decline in electricity load following natural disaster exposure, whereas negative interaction coefficients indicate that they amplify it. Standard errors are clustered at the MSA level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

### XI.VII.2 Alternative Measures of Natural Disaster Intensity

In the baseline analysis, I use the average wind speed to capture natural disaster intensity at the MSA-month level. As a robustness check, I instead define disaster intensity using the maximum monthly wind intensity observed within each MSA, although this measure may be more sensitive to the timing of a single extreme wind event within a month. Table **??** shows that the main results remain qualitatively similar under this alternative definition. For Category 4+ hurricanes defined using the maximum wind speed measure, both the average treatment effect and the estimated interaction coefficients closely resemble the baseline estimates, indicating that the findings are robust to alternative measures of disaster intensity.

Overall, these results suggest that the baseline average-intensity measure provides a more conservative definition of disaster exposure by requiring severe winds to be more broadly experienced within an MSA. For the average wind intensity of an MSA to exceed a given hurricane-category threshold, severe winds must affect a sufficiently large portion of the MSA. In contrast, the maximum-intensity measure classifies an MSA based on the strongest wind observed at any single location on any day within a month, even if such extreme winds are highly localized. Consistent with this distinction, the baseline Category 2+ results most closely resemble those obtained under the maximum-intensity Category 4+ definition. This similarity suggests that the two definitions identify disaster exposures of broadly comparable severity, despite differing substantially in how the spatial extent of exposure enters their construction.

Table 24: Robustness to an Alternative Disaster Intensity Measure: Maximum Intensity

| | | Change in Monthly MSA Electricity Load |
|---|---|---|
| | | Maximum Intensity: Category 4+ |
| *Panel A: Average Treatment Effect* | | |
| | Average treatment effect | $-0.0388^{***}$ (0.0032) |
| *Panel B: Interaction Coefficients* | | |
| Labor Market | Industrial Composition ($HHI$) | $-1.593^{***}$ (0.224) |
| Road Network | Road Connectivity ($wMEBC$) | $3.116^{***}$ (0.779) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.1252^{***}$ (0.0174) |
| Demographics | Population Density | $-0.000285^{***}$ (0.0000569) |
| | Observations | 30,263 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of the urban-structure characteristics and population density. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. As an alternative to the baseline disaster measure based on average MSA wind intensity, disaster exposure is defined here using whether the maximum monthly wind intensity within the MSA reaches Category 4 or above. The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Standard errors are clustered at the MSA level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

### XI.VII.3 Alternative LP-DiD Specification

**Controlling for Pre-Period Exposure** In the baseline LP-DiD specification, I implement the LP–DiD specification and show that treated and control MSAs do not systematically differ in their exposure to natural disasters prior to treatment. More specifically, Table 4 shows no statistically significant differences between the two groups in either average or maximum pre-treatment wind intensity.

As a robustness check, I augment the baseline specification with controls for prior disaster exposure, measured using either average or maximum pre-treatment wind intensity. Table 25 shows that the estimated treatment effects and interaction coefficients remain highly consistent with the baseline findings. These results indicate that both the average treatment effect and the estimated heterogeneity across urban structures are robust to explicitly controlling for differences in pre-period natural disaster exposure across MSAs.

Table 25: Robustness to Pre-period Disaster Intensity Controls

| | | Change in Monthly MSA Electricity Load | |
|---|---|---|---|
| | | Additional Control Variable | |
| | | Avg. Pre-period Intensity | Max. Pre-period Intensity |
| *Panel A: Average Treatment Effect* | | | |
| | Average treatment effect | $-0.0292^{***}$ (0.0056) | $-0.0290^{***}$ (0.0056) |
| *Panel B: Interaction Coefficients* | | | |
| Labor Market | Industrial Composition ($HHI$) | $-1.591^{***}$ (0.288) | $-1.583^{***}$ (0.288) |
| Road Network | Road Connectivity ($wMEBC$) | $3.026^{***}$ (0.951) | $3.048^{***}$ (0.936) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.0747^{***}$ (0.0279) | $0.0744^{***}$ (0.0275) |
| Demographics | Population Density | $-0.000423^{***}$ (0.0000879) | $-0.000422^{***}$ (0.0000870) |
| | Observations | 30,138 | 30,138 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of the urban-structure characteristics and population density. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. The first column controls for prior disaster exposure using average pre-period wind intensity, while the second column uses maximum pre-period wind intensity. The specification additionally controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the estimated decline in electricity load following natural disaster exposure, whereas negative interaction coefficients indicate that they amplify it. Standard errors are clustered at the MSA level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

**Longer Stabilization Period** In the baseline LP–DiD specification, I also impose a one-month stabilization period prior to treatment onset. This restriction reflects the assumption that the effects of natural disasters are relatively short-lived, such that requiring cities to remain untreated for two consecutive months before a new disaster occurs is sufficient to ensure that they have returned to a clean untreated state. Nevertheless, this assumption may be restrictive if the effects of prior disaster exposure persist for longer, such that an MSA requires a longer untreated period before it can reasonably be considered to have returned to a clean untreated state. To examine the sensitivity of the results to this assumption, I extend the stabilization period by requiring no changes in status during the preceding $L$ periods before treatment entry, thereby imposing a more conservative definition of a clean untreated state.

Table 26 examines the robustness of the baseline specification by extending the stabilization period from one month to between two and six months. Longer stabilization periods impose progressively stricter pre-treatment clean-window requirements by excluding observations that experience additional natural disaster exposure over increasingly longer pre-treatment intervals. Across all specifications, the estimated treatment and interaction effects remain highly stable in both magnitude and statistical significance, while the number of observations declines only modestly as the stabilization period increases.

Table 26: Sensitivity to the Stabilization Length

| | | Stabilization Length ($s$) | | | | |
|---|---|---|---|---|---|---|
| | | $s = 2$ | $s = 3$ | $s = 4$ | $s = 5$ | $s = 6$ |
| *Panel A: Average Treatment Effect* | | | | | | |
| | Average treatment effect | $-0.0288^{***}$ (0.0056) | $-0.0288^{***}$ (0.0056) | $-0.0288^{***}$ (0.0056) | $-0.0288^{***}$ (0.0056) | $-0.0288^{***}$ (0.0056) |
| *Panel B: Interaction Coefficients* | | | | | | |
| Labor Market | Industrial Composition ($HHI$) | $-1.553^{***}$ (0.301) | $-1.550^{***}$ (0.301) | $-1.549^{***}$ (0.301) | $-1.548^{***}$ (0.301) | $-1.548^{***}$ (0.301) |
| Road Network | Road Connectivity ($wMEBC$) | $3.038^{***}$ (0.925) | $3.037^{***}$ (0.926) | $3.040^{***}$ (0.927) | $3.039^{***}$ (0.925) | $3.042^{***}$ (0.925) |
| Trade Flow | Intra-city Goods Sourcing ($IntraTFR$) | $0.0652^{**}$ (0.0284) | $0.0652^{**}$ (0.0284) | $0.0652^{**}$ (0.0284) | $0.0651^{**}$ (0.0284) | $0.0652^{**}$ (0.0284) |
| Demographics | Population Density | $-0.000396^{***}$ (0.0000930) | $-0.000395^{***}$ (0.0000932) | $-0.000395^{***}$ (0.0000934) | $-0.000395^{***}$ (0.0000934) | $-0.000395^{***}$ (0.0000934) |
| | Observations | 30,188 | 30,158 | 30,128 | 30,098 | 30,068 |

*Notes:* Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of the urban-structure characteristics and population density. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Each column varies the stabilization length $s$, defined as the number of consecutive pre-event months during which control MSAs are required to exhibit no treatment-status transitions. The specification controls for heating degree days, cooling degree days, and electricity outages in both the event month and the preceding month, as well as the share of the housing stock built before 1980. Event-month and preceding-month time fixed effects are included. Standard errors are clustered at the MSA level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.